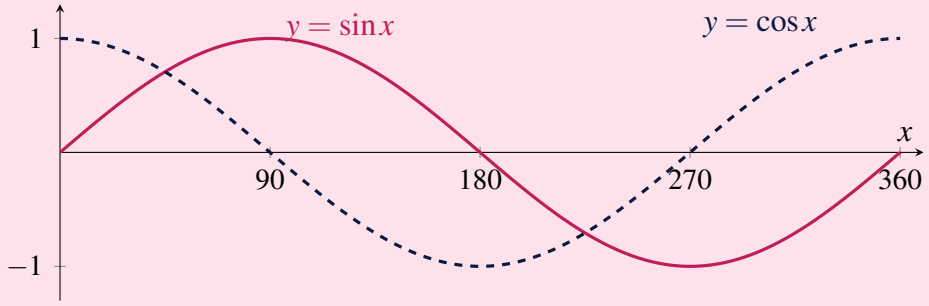
Advanced Trigonometry

Notes · Questions · Worked Answers

Pythagoras & right-angled trig · Exact values · Trig graphs & equations (0–180°, 0–360°) · Area $\frac{1}{2}ab\sin C$ · Sine rule & the ambiguous case · Cosine rule · Bearings · 3D trigonometry



How to use this booklet

- Read the pink **Key Notes** pages first.
- The questions are grouped by type into 12 sections, in syllabus order.
- Each white **Question** page is followed straight away by a yellow **Answer** page with full working — attempt every question before turning over.
- Work in the space under each question. Calculator allowed unless stated; give answers to 3 significant figures and angles to 1 decimal place.

Key Notes

0. Warm-up: Pythagoras and SOH CAH TOA

- Triangle $= \frac{1}{2} \times b \times h$ (h perpendicular to b)
- Pythagoras: $a^2 + b^2 = c^2$ (c = hypotenuse)
- 3D: diagonal of cuboid $= \sqrt{a^2 + b^2 + c^2}$

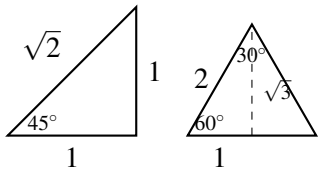
$$\sin\theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos\theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan\theta = \frac{\text{opp}}{\text{adj}}$$

Shortest distance from a point to a line is the **perpendicular** distance.

Angle of elevation / depression is measured from the horizontal.

1. Exact values (no calculator) — syllabus: PPT

θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined



Memory trick: sin of $0, 30, 45, 60, 90$ is $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; cos is the same list backwards; $\tan = \frac{\sin}{\cos}$.

2. Trig graphs and solving equations (0–180° and 0–360°)

- $y = \sin x$, $y = \cos x$: period $360°$, values between -1 and 1.
- $y = \tan x$: period $180°$, asymptotes at $x = 90°, 270°$.
- **Symmetry rules** (find the calculator value first):
  $\sin x$: x, $180 - x$
  $\cos x$: x, $360 - x$
  $\tan x$: x, $x + 180$
- $\sin x < 0 \Rightarrow$ answers in 180–360. $\cos x < 0 \Rightarrow$ answers in 90–270.

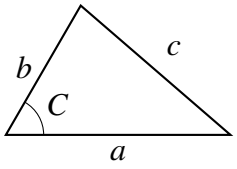
Example: $\sin x = -0.4$, $0 \le x \le 360$.
Calc: $\sin^{-1}(0.4) = 23.6°$ (reference angle)
$\Rightarrow x = 180 + 23.6 = 203.6°$
$x = 360 - 23.6 = 336.4°$

CAST: in quadrant 1 **A**ll positive, 2 **S**in, 3 **T**an, 4 **C**os.

For 0–180 only: sin gives **two** answers (if $0 < k < 1$), cos and tan give **one**.

3. Area of a triangle

$$\text{Area} = \frac{1}{2}ab\sin C \quad \text{(two sides and the angle \textbf{between} them)}$$

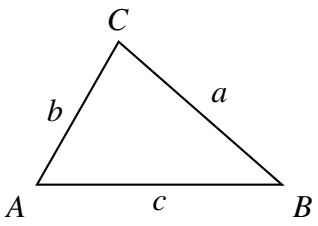


- Parallelogram $= ab\sin\theta$ (rhombus: $a^2\sin\theta$).
- Regular n-gon in circle radius r: $n \times \frac{1}{2}r^2\sin\left(\frac{360}{n}\right)$.
- Quadrilateral: split along a diagonal into two triangles.
- Reverse: $\sin C = \dfrac{2 \times \text{Area}}{ab}$ — remember C may be acute **or** $180 - C$.

4. Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \qquad \text{or} \qquad \frac{\sin A}{a} = \frac{\sin B}{b}$$

- Use when you know a **side and its opposite angle** (a matching pair) plus one more side or angle.
- Finding a side: put sides on top. Finding an angle: put angles on top.
- Angles in a triangle add to $180°$ — use it to find the third angle first.



5. Sine rule — the ambiguous case

When finding an **angle** with the sine rule, $\sin^{-1}$ gives one acute answer θ, but $180° - \theta$ has the **same sine**.

- Given angle A and sides a, b (angle not between them): if $a < b$ there may be **two** triangles.
- Check: is $180 - \theta +$ the given angle $< 180°$? If yes, both are possible.
- If $\sin\theta > 1 \Rightarrow$ **no triangle** exists. If $\sin\theta = 1 \Rightarrow$ exactly one (right-angled) triangle.
- If the question says "obtuse angle", give $180 - \theta$.

6. Cosine rule

$$a^2 = b^2 + c^2 - 2bc\cos A \qquad \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

- Side: two sides & the **included** angle (SAS).
- Angle: all three sides (SSS). Largest angle is opposite the longest side.
- Negative $\cos A \Rightarrow A$ is obtuse (the cosine rule is **never** ambiguous).

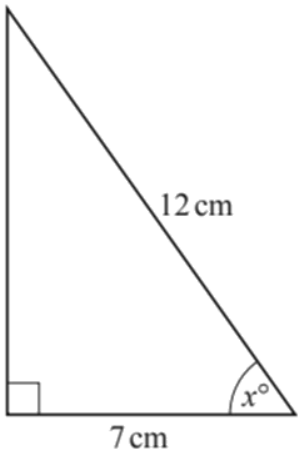
Which rule?
Matching pair (side + opposite angle)? → **Sine rule**.
SAS or SSS? → **Cosine rule**.
Right angle? → SOH CAH TOA / Pythagoras.

7. Mixed problems: bearings and 3D

Bearings: measured **clockwise from North**, 3 figures.
Back bearing = bearing $\pm 180°$.
Draw North lines at every point; use co-interior angles (add to $180°$) between parallel North lines.

3D: find a right-angled triangle that contains the line you need.
Cuboid diagonal $= \sqrt{a^2 + b^2 + c^2}$.
Angle between a line and a plane: drop a perpendicular to the plane, use the line, its projection and the height.

1

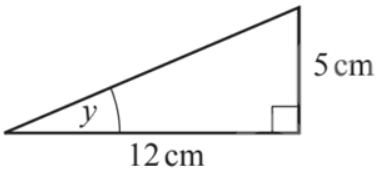


NOT TO SCALE

Calculate the value of x.

$x =$ ..... [2]

2 Find, as a fraction, the value of $\sin y$.



NOT TO SCALE

$\sin y =$ ..... [3]

3 The lengths of the sides of a triangle are 3 cm, 4 cm and 5 cm.

Find the sine of the smallest angle.

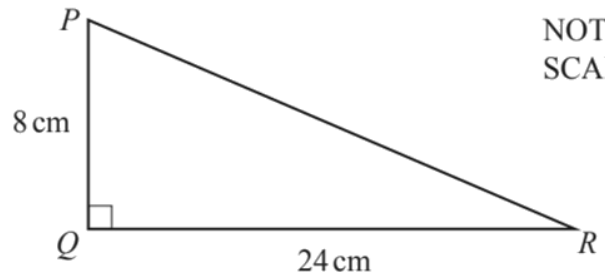
..... [1]

1 $\cos x = \dfrac{7}{12} \Rightarrow x = \cos^{-1}(0.5833) = \boxed{54.3^\circ}$

2 Hypotenuse $= \sqrt{12^2 + 5^2} = 13$. $\sin y = \dfrac{\text{opp}}{\text{hyp}} = \boxed{\dfrac{5}{13}} \Rightarrow y = \sin^{-1}\left(\frac{5}{13}\right) = \boxed{22.6^\circ}$

3 $3^2 + 4^2 = 5^2$, so the triangle is right-angled; the smallest angle is opposite 3: $\sin = \boxed{\frac{3}{5}} \Rightarrow$ angle $= \sin^{-1}(0.6) = \boxed{36.9^\circ}$

4



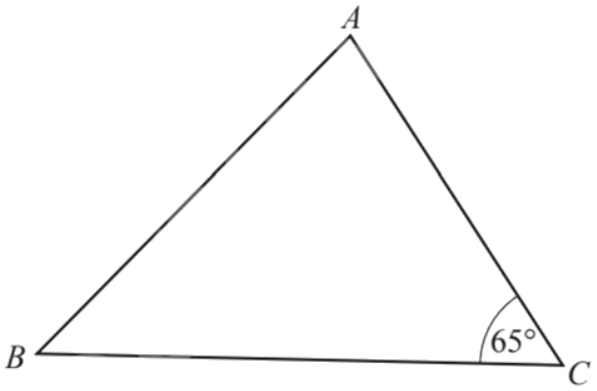
(i) Calculate the area of triangle PQR.

..... cm^2 [2]

(ii) Calculate angle PRQ.

Angle PRQ = ... [2]

5



The shortest distance from B to AC is 12.8 cm.

Calculate BC.

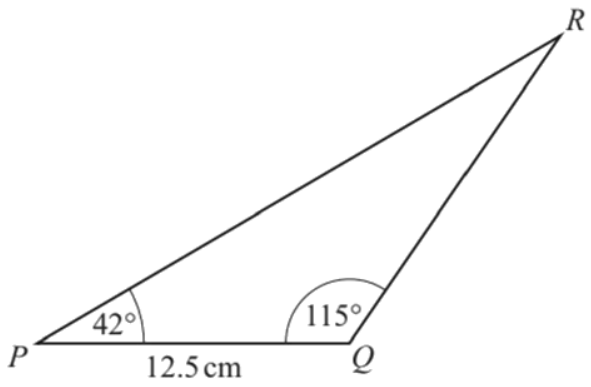
BC = ... cm [3]

4 (i) $\frac{1}{2} \times 8 \times 24 = \boxed{96 \text{ cm}^2}$ (ii) $\tan PRQ = \dfrac{8}{24} \Rightarrow \boxed{18.4^\circ}$

5 The perpendicular from B to AC makes a right-angled triangle with hypotenuse BC:

$$\sin 65^\circ = \frac{12.8}{BC} \Rightarrow BC = \frac{12.8}{\sin 65^\circ} = \boxed{14.1 \text{ cm}}$$

6



NOT TO SCALE

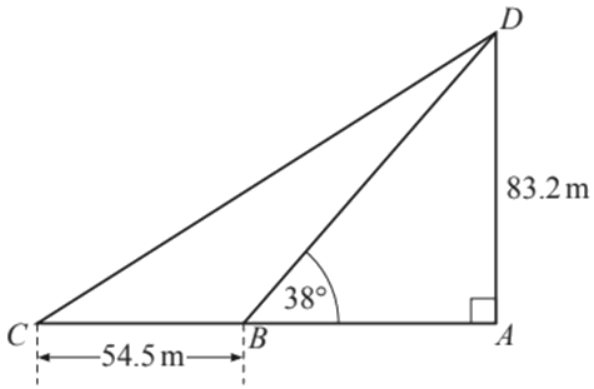
The diagram shows triangle PQR.

Calculate the shortest distance from Q to PR.

..... cm [3]

6 Shortest distance is the perpendicular from Q to PR: $d = PQ\sin 42^\circ = 12.5\sin 42^\circ =$ $\boxed{8.36 \text{ cm}}$

7



NOT TO SCALE

ACD is a right-angled triangle.
B is on AC and $BC = 54.5$ m.
$AD = 83.2$ m and angle $ABD = 38°$.

Calculate angle ACD.

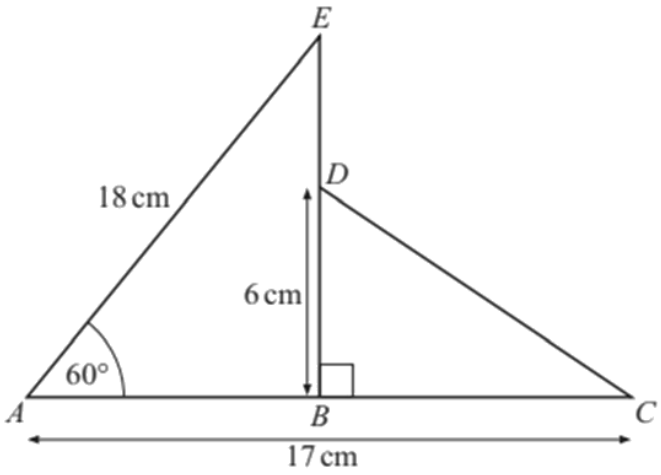
Angle ACD = ... [5]

7 In triangle ABD: $AB = \dfrac{83.2}{\tan 38^\circ} = 106.49$ m. $AC = 106.49 + 54.5 = 160.99$ m.

$\tan ACD = \dfrac{83.2}{160.99} \Rightarrow \boxed{\angle ACD = 27.3^\circ}$

8

NOT TO SCALE



The quadrilateral $ACDE$ is formed by two right-angled triangles ABE and BCD.
$AC = 17$ cm, $AE = 18$ cm and $BD = 6$ cm.

(a) Show that $CD = 10$ cm.

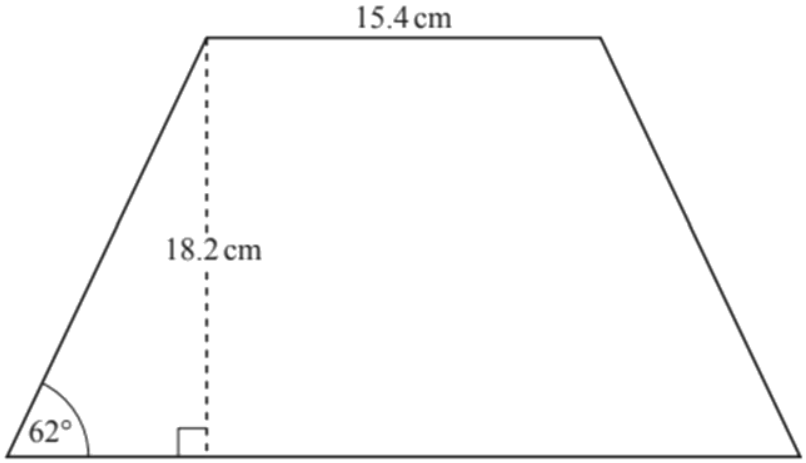
[5]

(b) Find the perimeter of the quadrilateral $ACDE$.
Give your answer in the form $p + k\sqrt{q}$.

..... cm [4]

8 (a) $AB = 18\cos 60° = 9$, so $BC = 17 - 9 = 8$. $CD = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$ ✓
(b) $BE = 18\sin 60° = 9\sqrt{3}$, so $DE = 9\sqrt{3} - 6$.
Perimeter $= AC + CD + DE + EA = 17 + 10 + (9\sqrt{3} - 6) + 18 = \boxed{39 + 9\sqrt{3}\text{ cm}}$

9



NOT TO SCALE

The diagram shows a trapezium.
The trapezium has one line of symmetry.

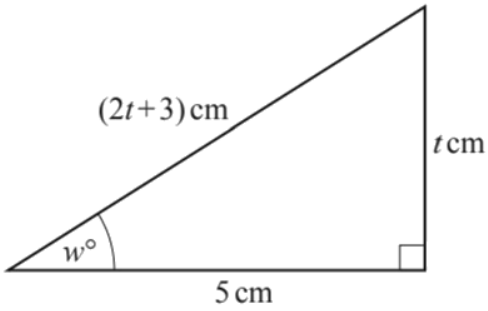
Work out the area of the trapezium.

..... cm^2 [4]

9 Overhang each side $= \dfrac{18.2}{\tan 62^\circ} = 9.677$; bottom $= 15.4 + 2(9.677) = 34.75$.

Area $= \frac{1}{2}(15.4 + 34.75) \times 18.2 = \boxed{456\ \text{cm}^2}$

10



NOT TO SCALE

The diagram shows a right-angled triangle.

Find the value of w.

$w =$ ..... [7]

10 Pythagoras: $(2t+3)^2 = t^2 + 5^2 \Rightarrow 4t^2 + 12t + 9 = t^2 + 25 \Rightarrow 3t^2 + 12t - 16 = 0$

$t = \dfrac{-12 + \sqrt{144 + 192}}{6} = 1.055$ (positive root). $\quad \tan w = \dfrac{t}{5} = 0.2110 \Rightarrow \boxed{w = 11.9}$

11 Complete the table using exact values. Do not use a calculator.

θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$					
$\cos\theta$					
$\tan\theta$					

[5]

12 Without using a calculator, find the exact value of

(a) $\sin 30° + \cos 60°$ **(b)** $\tan 45° \times \sin 60°$ **(c)** $\sin^2 45° + \cos^2 30°$

(d) $\dfrac{\tan 60°}{\cos 30°}$ **(e)** $2\sin 60° \cos 60°$

(a) ..... (b) ..... (c) ..... (d) ..... (e) ..... [5]

13 Find the acute angle x in each case.

(a) $\sin x = \dfrac{\sqrt{2}}{2}$ **(b)** $\tan x = \dfrac{1}{\sqrt{3}}$ **(c)** $2\cos x = 1$

[3]

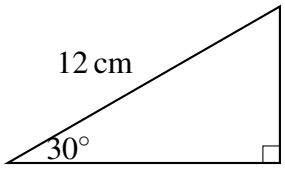
11

θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

12 (a) $\frac{1}{2}+\frac{1}{2}=\boxed{1}$ (b) $1\times\frac{\sqrt{3}}{2}=\boxed{\frac{\sqrt{3}}{2}}$ (c) $\frac{2}{4}+\frac{3}{4}=\boxed{\frac{5}{4}}$ (d) $\sqrt{3}\div\frac{\sqrt{3}}{2}=\boxed{2}$ (e) $2\times\frac{\sqrt{3}}{2}\times\frac{1}{2}=\boxed{\frac{\sqrt{3}}{2}}$

13 (a) $\boxed{45°}$ (b) $\frac{1}{\sqrt{3}}=\tan 30°\Rightarrow\boxed{30°}$ (c) $\cos x=\frac{1}{2}\Rightarrow\boxed{60°}$

14 **(a)** A right-angled triangle has hypotenuse 12 cm and one angle of 30°.
Find the exact lengths of the other two sides.



NOT TO SCALE

(b) Another right-angled triangle has an angle of 60° whose adjacent side is 5 cm. Find the exact length of the opposite side and of the hypotenuse.

[4]

15 An equilateral triangle has sides of length 6 cm.
(a) Show that its height is exactly $3\sqrt{3}$ cm.
(b) Find its exact area.

cm^2 ..... [4]

16 A triangle has sides 8 cm and 10 cm with an included angle of 60°.
Find the exact area of the triangle, giving your answer in the form $k\sqrt{3}$.

cm^2 ..... [3]

14 (a) opposite $= 12\sin 30° = \boxed{6 \text{ cm}}$; adjacent $= 12\cos 30° = \boxed{6\sqrt{3} \text{ cm}}$

(b) opposite $= 5\tan 60° = \boxed{5\sqrt{3} \text{ cm}}$; hypotenuse $= \dfrac{5}{\cos 60°} = \boxed{10 \text{ cm}}$

15 (a) Half-triangle: hypotenuse 6, base 3; height $= \sqrt{36-9} = \sqrt{27} = 3\sqrt{3}$ ✓

(b) $\frac{1}{2} \times 6 \times 3\sqrt{3} = \boxed{9\sqrt{3} \text{ cm}^2}$

16 $\frac{1}{2} \times 8 \times 10 \times \sin 60° = 40 \times \frac{\sqrt{3}}{2} = \boxed{20\sqrt{3} \text{ cm}^2}$

17 A right-angled isosceles triangle has shorter sides of length 7 cm.

Find the exact length of the hypotenuse and use the triangle to show that $\cos 45^\circ = \frac{\sqrt{2}}{2}$.

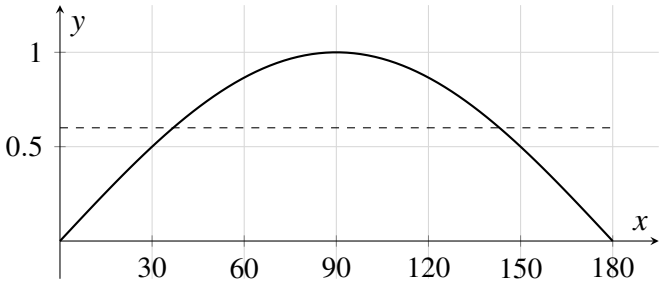
cm ..... [2]

17 Hypotenuse $= \sqrt{49+49} = \boxed{7\sqrt{2}\text{ cm}}$; $\cos 45^\circ = \dfrac{7}{7\sqrt{2}} = \dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$ ✓

18 The graph of $y = \sin x$ for $0° \leq x \leq 180°$ is shown, with the line $y = 0.6$.

(a) How many solutions does $\sin x = 0.6$ have in this range?

(b) One solution is $x = 36.9°$.
Use the symmetry of the graph to find the other.



(a) ..... (b) $x =$ ..... [3]

19 Solve each equation for $0° \leq x \leq 180°$. Give your answers correct to 1 decimal place.

(a) $\sin x = 0.35$ **(b)** $\cos x = -0.42$ **(c)** $\tan x = -2.5$

(d) $4\sin x - 1 = 2$ **(e)** $5\cos x + 2 = 0$ **(f)** $3\tan x = 7$

[10]

20 Write down the coordinates of the maximum point of $y = \sin x$ and of the minimum point of $y = \cos x$ for $0° \leq x \leq 180°$.

max (...,...) min ..... [2]

18 (a) The line cuts the curve twice: $\boxed{2}$. (b) Symmetry about $x = 90°$: $180 - 36.9 = \boxed{143.1°}$

19 (a) $\sin^{-1} 0.35 = 20.5°$; $180 - 20.5$: $\boxed{20.5°,\ 159.5°}$ (b) $\cos^{-1}(-0.42) = \boxed{114.8°}$
(c) $\tan^{-1}(-2.5) = -68.2°$; $+180$: $\boxed{111.8°}$ (d) $\sin x = 0.75$: $\boxed{48.6°,\ 131.4°}$
(e) $\cos x = -0.4$: $\boxed{113.6°}$ (f) $\tan x = \frac{7}{3}$: $\boxed{66.8°}$ (next is $246.8°$, out of range)

20 Max of $\sin x$: $\boxed{(90°, 1)}$; min of $\cos x$: $\boxed{(180°, -1)}$

21 **(a)** Given that $\sin 52° = 0.788$, find another angle between $0°$ and $180°$ whose sine is 0.788.

(b) Given that $\cos 70° = 0.342$, find the angle between $0°$ and $180°$ whose cosine is -0.342.

(a) ..... (b) ..... [2]

22 For $0° \leq x \leq 180°$, state the number of solutions of

(a) $\sin x = 0.4$ **(b)** $\sin x = 1$ **(c)** $\sin x = -0.2$ **(d)** $\cos x = -0.9$

(a) ..... (b) ..... (c) ..... (d) ..... [4]

23 Solve $\sin^2 x = 0.25$ for $0° \leq x \leq 180°$.

$x =$ ..... [3]

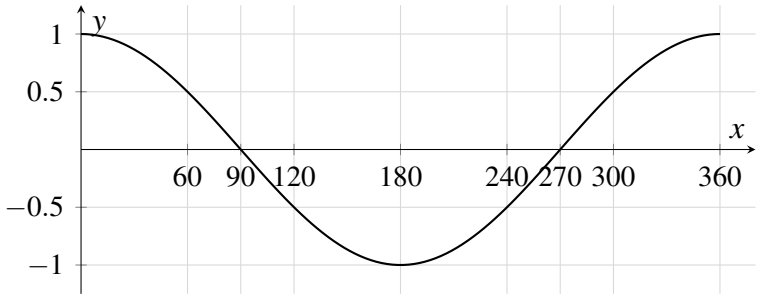
21 (a) $180 - 52 = \boxed{128^\circ}$ (b) $\cos(180-70) = -\cos 70$: $\boxed{110^\circ}$

22 (a) $\boxed{2}$ (b) $\boxed{1}$ ($x = 90^\circ$) (c) $\boxed{0}$ ($\sin x \ge 0$ here) (d) $\boxed{1}$

23 $\sin x = \pm 0.5$; in 0–180 only $\sin x = 0.5$ works: $\boxed{x = 30^\circ,\ 150^\circ}$

24 The graph of $y = \cos x$ for $0° \leq x \leq 360°$ is shown.

Use the graph to solve the equation $\cos x = 0.5$.



$x =$ ..... [2]

25 Solve for $0° \leq x \leq 360°$, giving answers to 1 decimal place where necessary.

(a) $\cos x = 0.3$ **(b)** $\tan x = 1.8$ **(c)** $\sin x = -0.72$

(d) $2\cos x + 1 = 0$ **(e)** $3\tan x + 4 = 0$

[10]

26 One solution of $\cos x = -0.8$ is $x = 143.1°$.
Find the other solution for $0° \leq x \leq 360°$.

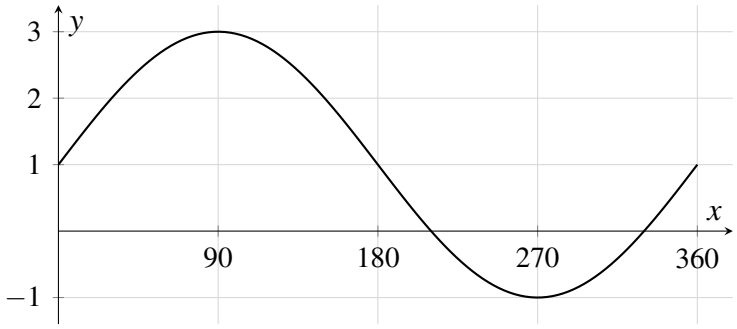
$x =$ ..... [1]

24 Line $y = 0.5$ meets the curve at $\boxed{x = 60° \text{ and } 300°}$

25 (a) $72.5°$, $360 - 72.5$: $\boxed{72.5°,\ 287.5°}$ (b) $60.9°$, $+180$: $\boxed{60.9°,\ 240.9°}$
(c) reference $\sin^{-1} 0.72 = 46.1°$: $\boxed{226.1°,\ 313.9°}$ (d) $\cos x = -\frac{1}{2}$: $\boxed{120°,\ 240°}$
(e) $\tan x = -\frac{4}{3}$, calc $-53.1°$: $\boxed{126.9°,\ 306.9°}$

26 $360 - 143.1 = \boxed{216.9°}$

27 The graph of $y = 2\sin x + 1$ for $0° \leq x \leq 360°$ is shown.



(a) Write down the maximum and minimum values of y and where they occur.

(b) Solve $2\sin x + 1 = 0$.

(b) $x =$ ..... [3]

28 **(a)** Write down the equations of the asymptotes of $y = \tan x$ for $0° \leq x \leq 360°$.

(b) State the period of $y = \tan x$ and the period of $y = \sin x$.

[3]

29 Solve $\tan^2 x = 3$ for $0° \leq x \leq 360°$.

$x =$ ..... [3]

30 Solve $\sin x = \cos x$ for $0° \leq x \leq 360°$.

$x =$ ..... [2]

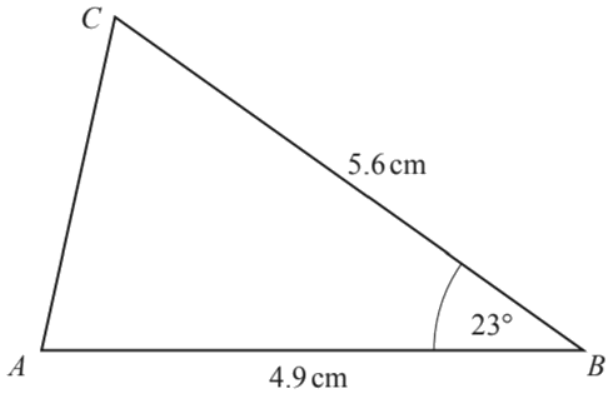
27 (a) Max $\boxed{3}$ at $x = 90^\circ$; min $\boxed{-1}$ at $x = 270^\circ$. (b) $\sin x = -\frac{1}{2}$: $\boxed{210^\circ,\ 330^\circ}$

28 (a) $\boxed{x = 90^\circ,\ x = 270^\circ}$ (b) tan: $\boxed{180^\circ}$; sin: $\boxed{360^\circ}$

29 $\tan x = \pm\sqrt{3}$: $\boxed{60^\circ,\ 120^\circ,\ 240^\circ,\ 300^\circ}$

30 Divide by $\cos x$: $\tan x = 1 \Rightarrow \boxed{45^\circ,\ 225^\circ}$

31

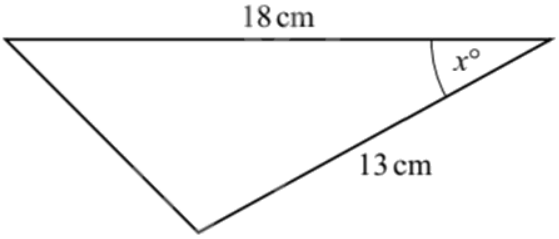


NOT TO SCALE

Calculate the area of triangle ABC.

..... cm^2 [2]

32



NOT TO SCALE

The area of the triangle is $50\,cm^2$.

Calculate the value of $\sin x$.

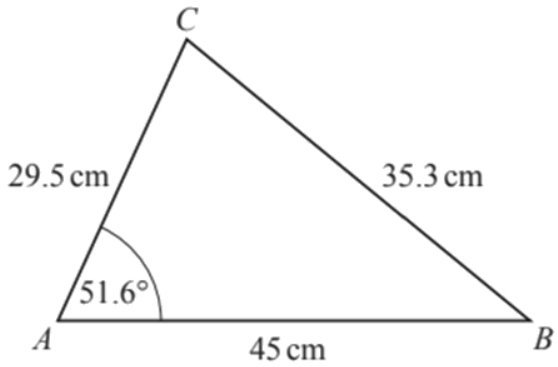
$\sin x =$ ..... [2]

31 Angle $23°$ is between 5.6 and 4.9: $\frac{1}{2} \times 4.9 \times 5.6 \times \sin 23° = \boxed{5.36 \text{ cm}^2}$

32 $50 = \frac{1}{2} \times 18 \times 13 \times \sin x = 117 \sin x \Rightarrow \sin x = \dfrac{50}{117} = \boxed{0.427}$

In degrees: $x = \sin^{-1}(0.4274) = \boxed{25.3°}$

33



NOT TO SCALE

In triangle ABC, $AB = 45$ cm, $AC = 29.5$ cm, $BC = 35.3$ cm and angle $CAB = 51.6°$.

(i) Calculate angle ABC.

Angle $ABC =$ ..... [3]

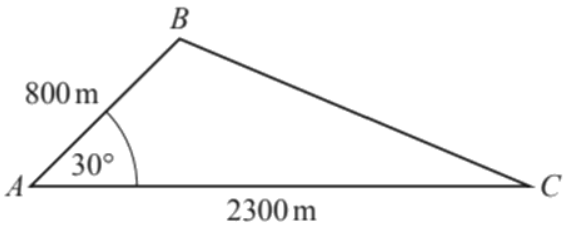
(ii) Calculate the area of triangle ABC.

..... cm^2 [2]

33 (i) Sine rule: $\dfrac{\sin B}{29.5} = \dfrac{\sin 51.6^\circ}{35.3} \Rightarrow \sin B = 0.6550 \Rightarrow \boxed{\angle ABC = 40.9^\circ}$

(ii) $\frac{1}{2} \times 45 \times 29.5 \times \sin 51.6^\circ = \boxed{520 \text{ cm}^2}$

34



NOT TO SCALE

The diagram shows some land in the shape of a triangle ABC.
Houses are built on this land.
Each house requires $400\,m^2$ of land.

Find the greatest number of houses that can be built on this land.

..... [3]

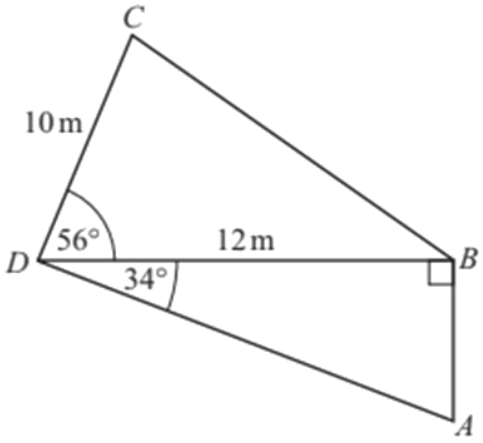
35 Find the area of a regular hexagon with side length 7.4 cm.

..... cm^2 [3]

34 Area $= \frac{1}{2} \times 800 \times 2300 \times \sin 30° = 460\,000$ m^2. $460\,000 \div 400 =$ $\boxed{1150 \text{ houses}}$

35 6 equilateral triangles: $6 \times \frac{1}{2} \times 7.4^2 \times \sin 60° =$ $\boxed{142 \text{ cm}^2}$

36



NOT TO SCALE

The diagram shows a quadrilateral $ABCD$.
$CD = 10$ m and $DB = 12$ m.
Angle $DBA = 90°$, angle $CDB = 56°$ and angle $ADB = 34°$.

(a) Calculate the length of AB.

$AB =$ ..... m [2]

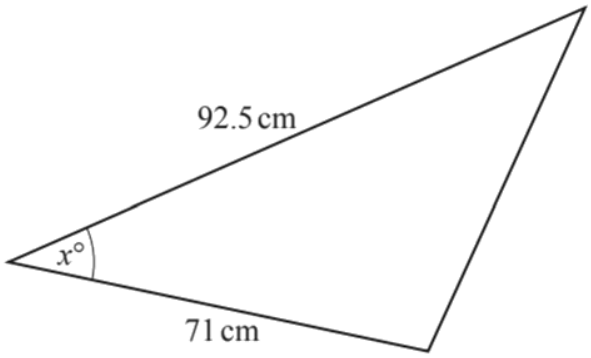
(b) Calculate the area of the quadrilateral $ABCD$.

..... m^2 [3]

36 (a) $\tan 34^\circ = \dfrac{AB}{12} \Rightarrow AB = 12\tan 34^\circ = \boxed{8.09\text{ m}}$

(b) $ABD = \frac{1}{2} \times 12 \times 8.094 = 48.56$; $BCD = \frac{1}{2} \times 10 \times 12 \times \sin 56^\circ = 49.74$. Total $= \boxed{98.3\text{ m}^2}$

37



NOT TO SCALE

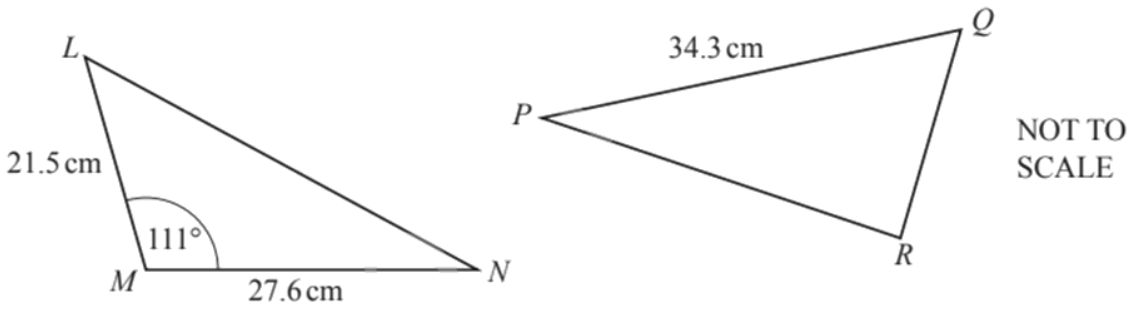
The diagram shows a triangle with an acute angle marked $x°$.
The area of the triangle is $2143\,\text{cm}^2$.

Work out the value of x.

$x =$ ..... [2]

37 $2143 = \frac{1}{2} \times 92.5 \times 71 \times \sin x \Rightarrow \sin x = \dfrac{4286}{6567.5} = 0.6526 \Rightarrow \boxed{x = 40.7}$

38



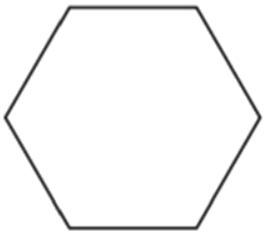
Triangle PQR has the same area as triangle LMN.

Calculate the shortest distance from R to the line PQ.

..... cm [3]

38 Area $LMN = \frac{1}{2} \times 21.5 \times 27.6 \times \sin 111^\circ = 277.0\ \text{cm}^2$.
Area $PQR = \frac{1}{2} \times PQ \times h \Rightarrow 277.0 = \frac{1}{2} \times 34.3 \times h \Rightarrow h = \boxed{16.2\ \text{cm}}$

39



NOT TO SCALE

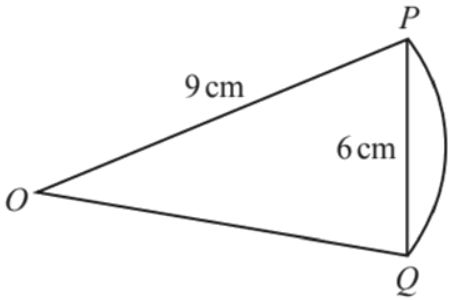
The diagram shows a regular hexagon.
The area of the hexagon is 127.3 cm^2.

(i) Show that the length of one side of the hexagon is 7.0 cm, correct to 1 decimal place.

[4]

39 $6 \times \frac{1}{2}s^2 \sin 60° = 127.3 \Rightarrow s^2 = \dfrac{127.3}{3\sin 60°} = 48.997 \Rightarrow s = 6.9998 = 7.0$ cm ✓

40



NOT TO SCALE

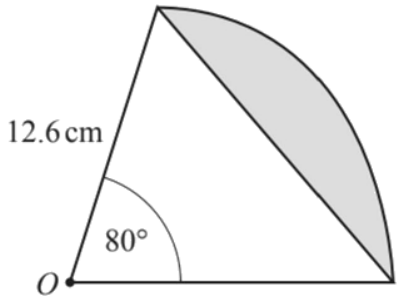
The diagram shows a sector of a circle with centre O and radius 9 cm.
The length of the chord PQ is 6 cm.

Calculate the length of the arc PQ.

..... cm [3]

40 Half the chord: $\sin\frac{\theta}{2} = \frac{3}{9} \Rightarrow \frac{\theta}{2} = 19.47^\circ \Rightarrow \theta = 38.94^\circ$
Arc $= \frac{38.94}{360} \times 2\pi \times 9 = \boxed{6.12 \text{ cm}}$

41



NOT TO SCALE

The diagram shows a sector of a circle, centre O, radius 12.6 cm.

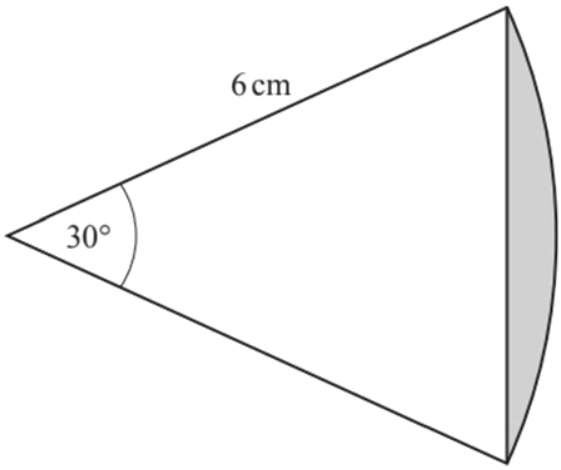
Calculate the perimeter of the shaded segment.

..... cm [4]

41 Arc $= \frac{80}{360} \times 2\pi \times 12.6 = 17.59$; $\text{chord}^2 = 2(12.6)^2 - 2(12.6)^2 \cos 80^\circ \Rightarrow$ chord $= 16.20$

Perimeter $= \boxed{33.8 \text{ cm}}$

42



NOT TO SCALE

The area of the shaded segment is $(a\pi + b)$ cm^2.

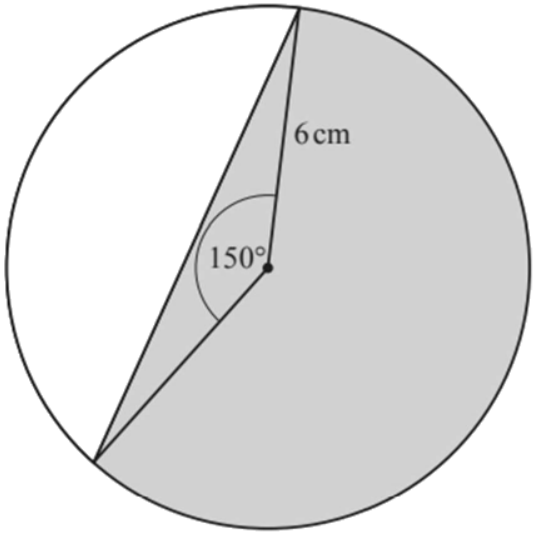
Find the value of a and the value of b.

$a =$ .....

$b =$ ..... [4]

42 Sector $= \frac{30}{360}\pi(36) = 3\pi$; triangle $= \frac{1}{2}(6)(6)\sin 30^\circ = 9$. Segment $= 3\pi - 9 \Rightarrow$ $\boxed{a = 3,\ b = -9}$

43



NOT TO SCALE

A sector of a circle with radius 6 cm has a sector angle of 150°.

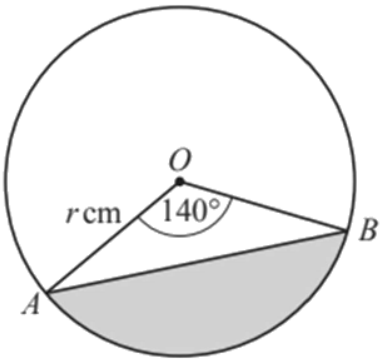
Find the exact value of the area of the shaded region.
Give your answer in its simplest form.

..... cm^2 [4]

43 Shaded = major sector (210°) + triangle.

Sector $= \frac{210}{360}\pi(6)^2 = 21\pi$; triangle $= \frac{1}{2}\times 6\times 6\times \sin 150^\circ = 9$. Shaded $= \boxed{21\pi + 9 \text{ cm}^2}$

44



NOT TO SCALE

A and B are points on a circle, centre O, radius r cm.
The area of the shaded segment is 65cm^2.

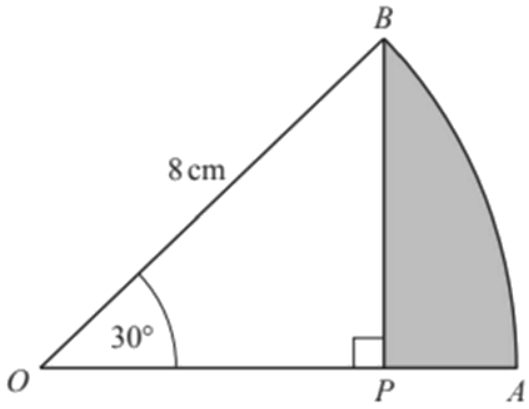
Calculate the value of r.

$r =$ ..... [4]

44 Segment $= \frac{140}{360}\pi r^2 - \frac{1}{2}r^2 \sin 140^\circ = 1.2217r^2 - 0.3214r^2 = 0.9003r^2$

$0.9003r^2 = 65 \Rightarrow r^2 = 72.19 \Rightarrow \boxed{r = 8.50}$

45



NOT TO SCALE

OAB is the sector of a circle, centre *O*.
OB = 8 cm and angle *AOB* = 30°.
BP is perpendicular to *OA*.

(a) Calculate *AP*.

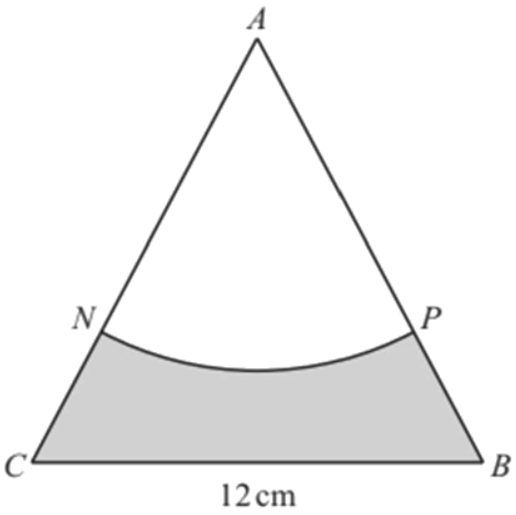
AP = ... cm [3]

(b) Work out the area of the shaded region *APB*.

..... cm^2 [3]

45 (a) $OP = 8\cos 30^\circ = 4\sqrt{3} = 6.928$; $AP = 8 - 6.928 = \boxed{1.07 \text{ cm}}$
(b) Sector $= \frac{30}{360}\pi(64) = 16.76$; triangle $OPB = \frac{1}{2} \times 6.928 \times 4 = 13.86$. Shaded $= \boxed{2.90 \text{ cm}^2}$

46



NOT TO SCALE

The diagram shows an equilateral triangle ABC with all sides of length 12 cm.
ANP is a sector of a circle, centre A.
N lies on AC such that $AN : NC = 2 : 1$.
P lies on AB such that $AP : PB = 2 : 1$.

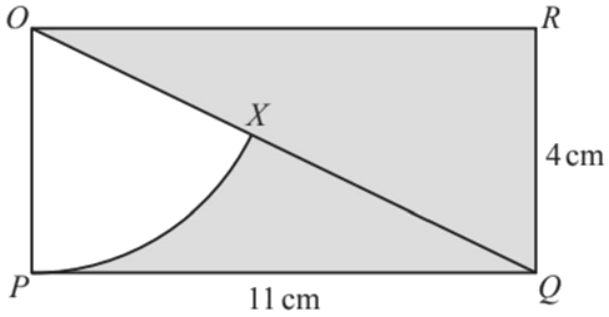
Calculate the area of the shaded region.

..... cm^2 [4]

46 $AN = AP = \frac{2}{3} \times 12 = 8$. Triangle $= \frac{1}{2}(12)(12)\sin 60^\circ = 62.35$; sector $= \frac{60}{360}\pi(8)^2 = 33.51$

Shaded $= 62.35 - 33.51 = \boxed{28.8 \text{ cm}^2}$

47



NOT TO SCALE

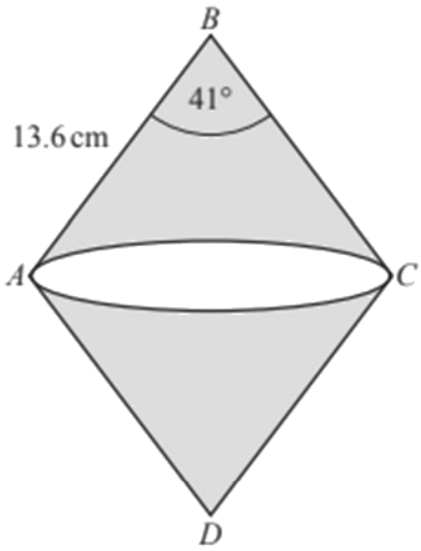
The diagram shows a rectangle $OPQR$ with length 11 cm and width 4 cm.
OQ is a diagonal and OPX is a sector of a circle, centre O.

Calculate the percentage of the rectangle that is shaded.

..... % [5]

47 $\tan POQ = \frac{11}{4} \Rightarrow \angle POQ = 70.02°$. Sector $OPX = \frac{70.02}{360}\pi(4)^2 = 9.78$
Shaded $= 44 - 9.78 = 34.22$; $\frac{34.22}{44} \times 100 = \boxed{77.8\%}$

48



NOT TO SCALE

ABCD is a rhombus with side length 13.6 cm.
Angle $ABC = 41°$.
BAC is a sector of a circle with centre *B*.
DAC is a sector of a circle with centre *D*.

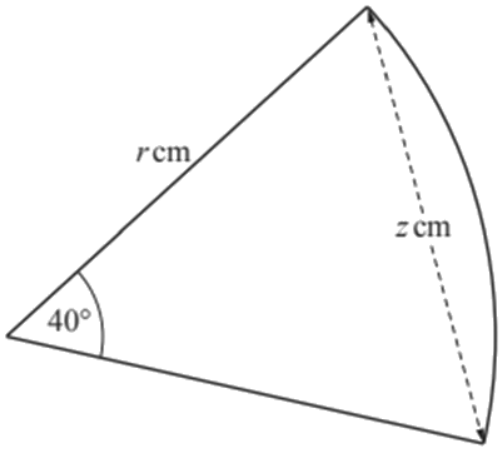
Calculate the shaded area.

..... cm^2 [4]

48 Triangle $ABC = \frac{1}{2}(13.6)^2 \sin 41^\circ = 60.67$; sector $= \frac{41}{360}\pi(13.6)^2 = 66.18$; segment $= 5.51$.

Shaded $= 2 \times$ triangle $- 2 \times$ segment $= 121.34 - 11.01 = \boxed{110 \text{ cm}^2}$

49



NOT TO SCALE

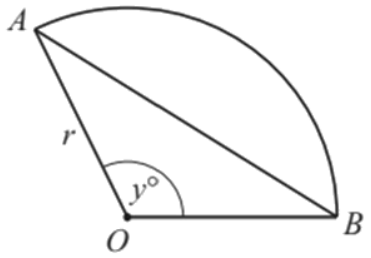
This sector of a circle has radius r and perimeter 20 cm.

Find the value of z.

$z =$ ..... [6]

49 $2r + \frac{40}{360} \times 2\pi r = 20 \Rightarrow r\left(2 + \frac{2\pi}{9}\right) = 20 \Rightarrow r = 7.4125$
$z^2 = r^2 + r^2 - 2r^2 \cos 40^\circ \Rightarrow \boxed{z = 5.07}$

50



NOT TO SCALE

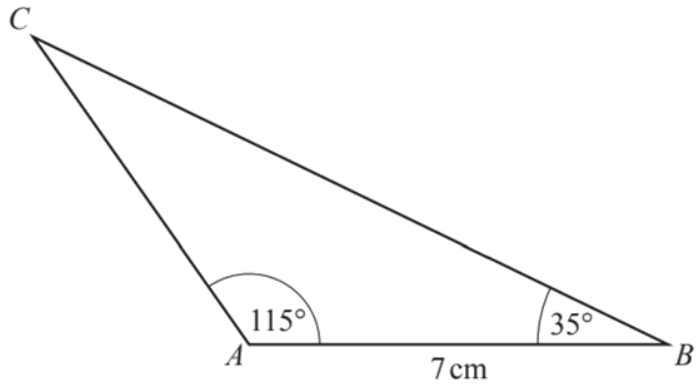
The diagram shows a sector, OAB, of a circle with centre O and radius r.
The area of triangle OAB is half the area of the sector.
Angle $AOB = y°$ and is obtuse.

(i) Show that $360 \sin y = \pi y$.

[2]

50 $\frac{1}{2}r^2 \sin y = \frac{1}{2} \times \frac{y}{360}\pi r^2 \Rightarrow \sin y = \frac{\pi y}{360} \Rightarrow 360 \sin y = \pi y$ ✓

51

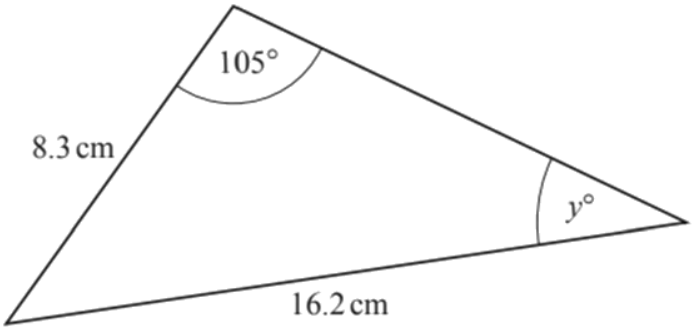


NOT TO SCALE

Calculate the length BC.

$BC =$ ..... cm [4]

52



NOT TO SCALE

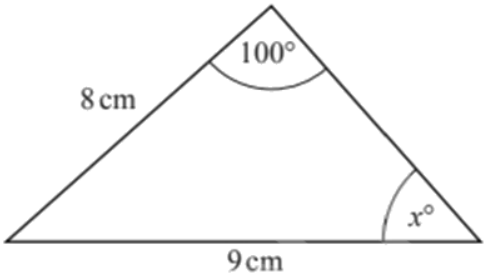
Calculate the value of y.

$y =$ ..... [3]

51 Angle $C = 180 - 115 - 35 = 30°$. $\dfrac{BC}{\sin 115°} = \dfrac{7}{\sin 30°} \Rightarrow BC = \boxed{12.7 \text{ cm}}$

52 $\dfrac{\sin y}{8.3} = \dfrac{\sin 105°}{16.2} \Rightarrow \sin y = 0.4949 \Rightarrow \boxed{y = 29.7}$

53



NOT TO SCALE

(a) Calculate the value of x.

$x =$ ..... [3]

(b) Calculate the area of the triangle.

..... cm^2 [3]

54 In triangle PQR, $PQ = 9$ cm, $QR = 13$ cm and angle $QPR = 72°$.
Calculate angle PRQ.

Angle $PRQ =$ ..... [3]

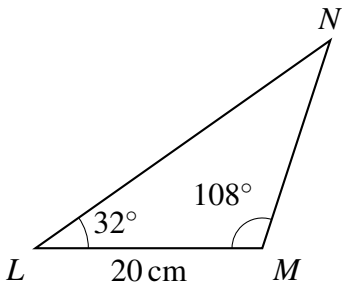
53 (a) $\dfrac{\sin x}{8} = \dfrac{\sin 100^\circ}{9} \Rightarrow \sin x = 0.8754 \Rightarrow \boxed{x = 61.1}$

(b) Third angle $= 180 - 100 - 61.1 = 18.9^\circ$ (between 8 and 9). Area $= \frac{1}{2}(8)(9)\sin 18.9^\circ =$ $\boxed{11.7 \text{ cm}^2}$

54 $\dfrac{\sin R}{9} = \dfrac{\sin 72^\circ}{13} \Rightarrow \sin R = 0.6584 \Rightarrow \boxed{41.2^\circ}$ (unique, since $9 < 13$)

55 In triangle LMN, $LM = 20\,\text{cm}$, angle $L = 32°$ and angle $M = 108°$.

(a) Find angle N.
(b) Calculate the length of the longest side.
(c) Calculate the length of the shortest side.



NOT TO SCALE

[5]

56 In triangle XYZ, $XY = 7.4\,\text{cm}$, angle $YXZ = 61°$ and angle $XZY = 53°$.
Calculate **(a)** YZ, **(b)** XZ.

(a) ..... cm (b) ..... [4]

57 Prove the sine rule $\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$ by drawing the perpendicular from C to AB and finding its length in two different ways.

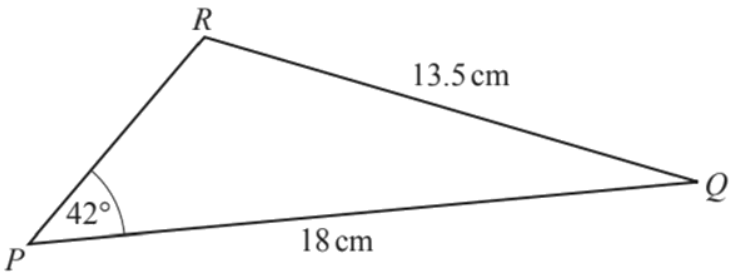
[3]

55 (a) $180 - 32 - 108 = \boxed{40^\circ}$ (b) opposite 108°: $LN = \dfrac{20\sin 108^\circ}{\sin 40^\circ} = \boxed{29.6\text{ cm}}$
(c) opposite 32°: $MN = \dfrac{20\sin 32^\circ}{\sin 40^\circ} = \boxed{16.5\text{ cm}}$

56 Angle $Y = 66^\circ$. (a) $YZ = \dfrac{7.4\sin 61^\circ}{\sin 53^\circ} = \boxed{8.10\text{ cm}}$ (b) $XZ = \dfrac{7.4\sin 66^\circ}{\sin 53^\circ} = \boxed{8.46\text{ cm}}$

57 Let the perpendicular be h. Triangle ACH: $h = b\sin A$; triangle BCH: $h = a\sin B$.
So $a\sin B = b\sin A \Rightarrow \dfrac{a}{\sin A} = \dfrac{b}{\sin B}$ ✓

58



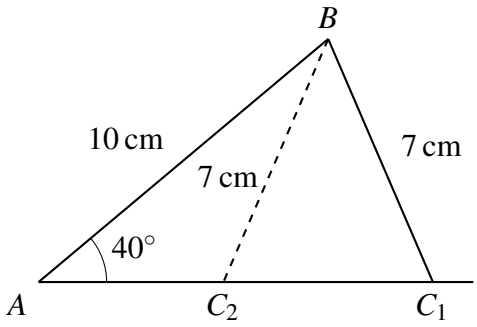
Calculate the obtuse angle PRQ.

Angle PRQ = ... [4]

59 In triangle ABC, $AB = 10$ cm, $BC = 7$ cm and angle $BAC = 40°$.
The diagram shows that two different triangles, ABC_1 and ABC_2, are possible.

(a) Find the two possible values of angle ACB.

(b) Find the two possible lengths of AC.



NOT TO SCALE

[6]

58 $\dfrac{\sin R}{18} = \dfrac{\sin 42^\circ}{13.5} \Rightarrow \sin R = 0.8922 \Rightarrow R = 63.1^\circ$ (acute). Obtuse: $180 - 63.1 = \boxed{116.9^\circ}$

59 (a) $\sin C = \dfrac{10\sin 40^\circ}{7} = 0.9183 \Rightarrow C = \boxed{66.7^\circ}$ or $\boxed{113.3^\circ}$ (both valid: $40 + 113.3 < 180$)

(b) $C = 66.7^\circ$: $B = 73.3^\circ$, $AC = \dfrac{7\sin 73.3^\circ}{\sin 40^\circ} = \boxed{10.4 \text{ cm}}$; $C = 113.3^\circ$: $B = 26.7^\circ$, $AC = \dfrac{7\sin 26.7^\circ}{\sin 40^\circ} = \boxed{4.89 \text{ cm}}$

60 In triangle PQR, $PQ = 16\,\text{cm}$, $QR = 12\,\text{cm}$ and angle $QPR = 38°$. Angle PRQ is obtuse.
Calculate angle PRQ.

Angle $PRQ =$ ..... [4]

61 A student tries to draw a triangle with $BC = 4\,\text{cm}$, $AC = 10\,\text{cm}$ and angle $BAC = 30°$.
Use the sine rule to show that no such triangle exists.

[2]

62 For each set of measurements, state how many different triangles ABC can be drawn. Show your working.

(a) $A = 30°$, $a = 5\,\text{cm}$, $b = 10\,\text{cm}$

(b) $A = 50°$, $a = 12\,\text{cm}$, $b = 9\,\text{cm}$

(c) $A = 35°$, $a = 6\,\text{cm}$, $b = 9\,\text{cm}$

(a) ..... (b) ..... (c) ..... [4]

63 Explain why finding an angle with the cosine rule never gives an ambiguous answer, but finding an angle with the sine rule sometimes does.

[2]

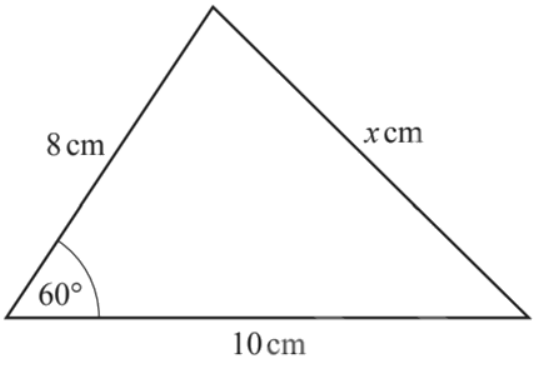
60 $\sin R = \dfrac{16\sin 38^\circ}{12} = 0.8209 \Rightarrow 55.2^\circ$; obtuse: $\boxed{124.8^\circ}$

61 $\sin B = \dfrac{10\sin 30^\circ}{4} = 1.25 > 1$, which is impossible, so **no triangle exists**.

62 (a) $\sin B = \dfrac{10\sin 30^\circ}{5} = 1 \Rightarrow B = 90^\circ$: $\boxed{1}$ (b) $\sin B = 0.575 \Rightarrow 35.1^\circ$ or 144.9°, but $50 + 144.9 > 180$: $\boxed{1}$
(c) $\sin B = 0.860 \Rightarrow 59.4^\circ$ or 120.6°, $35 + 120.6 < 180$: $\boxed{2}$

63 Each value of $\cos A$ gives exactly one angle between 0° and 180° (negative $\Rightarrow$ obtuse). But $\sin\theta = \sin(180^\circ - \theta)$, so one value of $\sin A$ can give two angles.

64



NOT TO SCALE

Find the value of x^2.

$x^2 =$ ..... [3]

65 A triangle has sides of length 11 cm, 10 cm and 9 cm.

Calculate the largest angle in the triangle.

..... [4]

64 $x^2 = 8^2 + 10^2 - 2(8)(10)\cos 60^\circ = 164 - 80 = \boxed{84}$

65 Largest angle opposite 11: $\cos\theta = \dfrac{10^2 + 9^2 - 11^2}{2(10)(9)} = \dfrac{60}{180} \Rightarrow \boxed{70.5^\circ}$

66 A triangle has sides of length 3 cm, 5 cm and 7 cm.

Work out the largest angle in the triangle.

..... [3]

67 A triangle has sides of length 8 cm and 11 cm with an included angle of 47°.
Calculate the length of the third side.

cm ..... [3]

68 In triangle DEF, $DE = 4.5$ m, $DF = 6.8$ m and angle $EDF = 132°$.
Calculate EF.

$EF =$ ..... m ..... [3]

66 Largest angle opposite 7: $\cos\theta = \dfrac{3^2+5^2-7^2}{2(3)(5)} = \dfrac{-15}{30} = -0.5 \Rightarrow \boxed{120^\circ}$

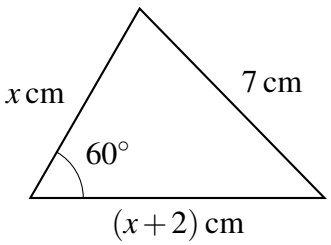
67 $x^2 = 8^2 + 11^2 - 2(8)(11)\cos 47^\circ = 64.97 \Rightarrow \boxed{8.06 \text{ cm}}$

68 $EF^2 = 20.25 + 46.24 - 61.2\cos 132^\circ = 107.44 \Rightarrow \boxed{10.4 \text{ m}}$ ($\cos 132^\circ < 0$, so add)

69 A triangle has sides 4 cm and 7 cm with an included angle of 120°.
Without using a calculator, find the exact length of the third side.

cm ..... [3]

70 In the triangle shown, two sides are x cm and $(x+2)$ cm, the angle between them is 60° and the third side is 7 cm.



NOT TO SCALE

(a) Show that $x^2 + 2x - 45 = 0$.

(b) Solve the equation to find x, correct to 2 decimal places.

(b) $x =$ ..... [5]

71 A parallelogram has sides of length 6 cm and 10 cm and an angle of 70°.
Calculate the lengths of both diagonals.

..... cm and ..... [4]

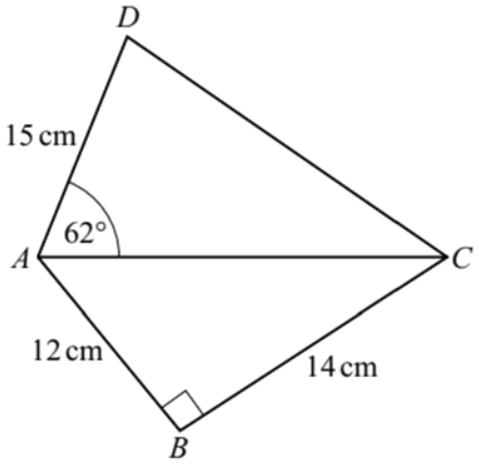
69 $x^2 = 16 + 49 - 56\cos 120^\circ = 65 + 28 = 93 \Rightarrow \boxed{\sqrt{93}\text{ cm}}$

70 (a) $49 = x^2 + (x+2)^2 - 2x(x+2)\cos 60^\circ = x^2 + 2x + 4 \Rightarrow x^2 + 2x - 45 = 0$ ✓

(b) $x = \dfrac{-2 + \sqrt{184}}{2} = \boxed{5.78}$

71 Angles 70° and 110°. $d_1^2 = 136 - 120\cos 70^\circ \Rightarrow \boxed{9.74\text{ cm}}$; $d_2^2 = 136 - 120\cos 110^\circ \Rightarrow \boxed{13.3\text{ cm}}$

72



NOT TO SCALE

The diagram shows a quadrilateral, $ABCD$, formed from two triangles, ABC and ACD.
ABC is a right-angled triangle.

(a) Calculate angle BAC.

Angle BAC = ... [2]

(b) Calculate BD.

BD = ... cm [4]

(c) Calculate the shortest distance from D to AC.

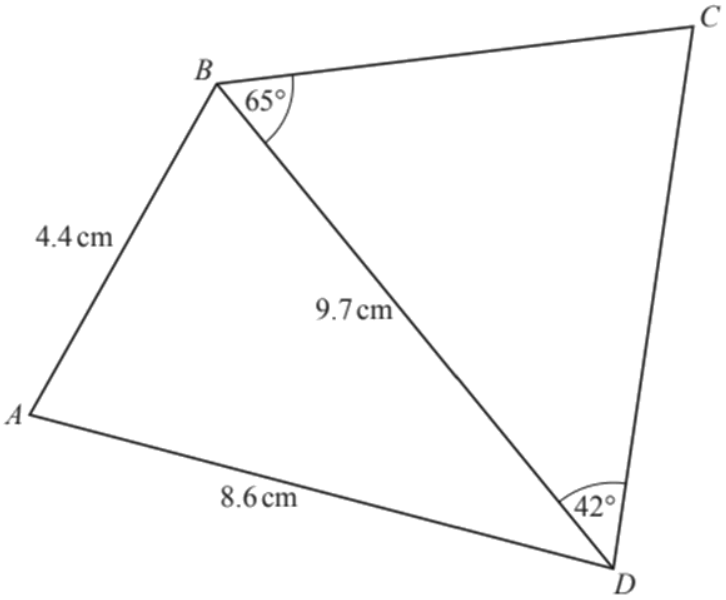
..... cm [3]

72 (a) $\cos BAC = \frac{12}{14} \Rightarrow \boxed{31.0^\circ}$

(b) $\angle BAD = 62 + 31.0 = 93.0^\circ$: $BD^2 = 12^2 + 15^2 - 2(12)(15)\cos 93.0^\circ \Rightarrow BD = \boxed{19.7 \text{ cm}}$

(c) $15 \sin 62^\circ = \boxed{13.2 \text{ cm}}$

73



NOT TO SCALE

(a) Calculate angle *ADB*.

Angle *ADB* = ... [3]

(b) Calculate *DC*.

DC = ... cm [4]

(c) Calculate the shortest distance from *C* to *BD*.

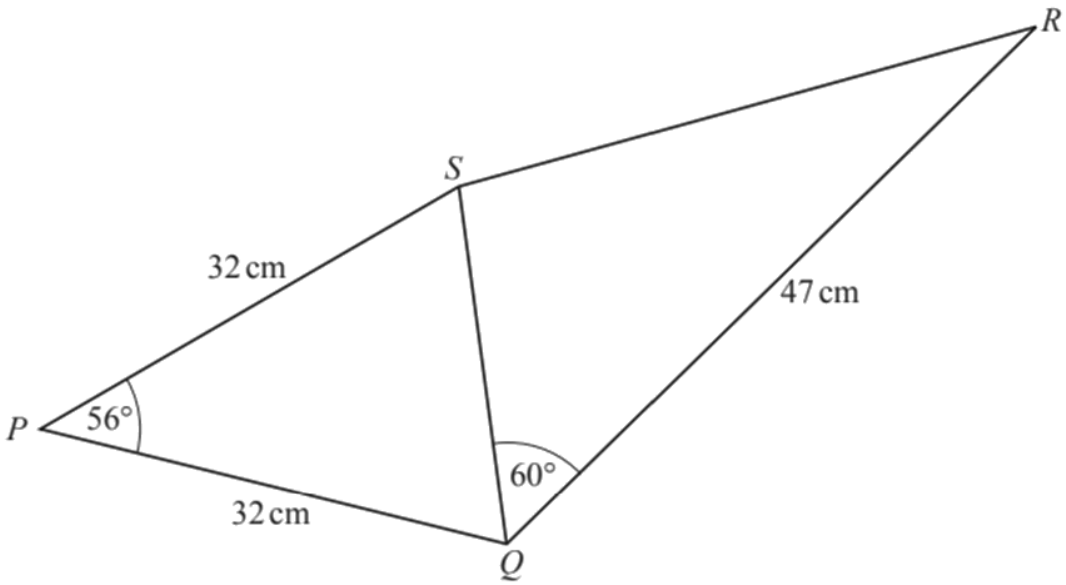
..... cm [3]

73 (a) $\cos ADB = \dfrac{8.6^2 + 9.7^2 - 4.4^2}{2(8.6)(9.7)} = 0.8913 \Rightarrow \boxed{27.0^\circ}$

(b) $\angle C = 180 - 65 - 42 = 73^\circ$: $DC = \dfrac{9.7 \sin 65^\circ}{\sin 73^\circ} = \boxed{9.19 \text{ cm}}$

(c) $9.19 \sin 42^\circ = \boxed{6.15 \text{ cm}}$

74



NOT TO SCALE

The diagram shows a quadrilateral $PQRS$ formed from two triangles, PQS and QRS.
Triangle PQS is isosceles, with $PQ = PS = 32$ cm and angle $SPQ = 56°$.
$QR = 47$ cm and angle $SQR = 60°$.

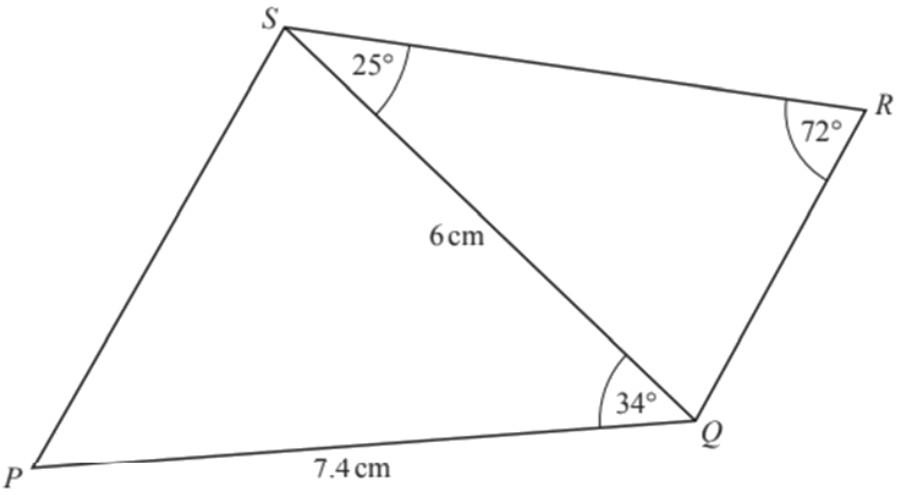
(i) Calculate SR.

$SR =$ ..... cm [4]

74 $\angle PQS = \angle PSQ = 62°$. $QS = \dfrac{32\sin 56°}{\sin 62°} = 30.05$

$SR^2 = 47^2 + 30.05^2 - 2(47)(30.05)\cos 60° \Rightarrow SR = \boxed{41.2 \text{ cm}}$

75



NOT TO SCALE

The diagram shows a quadrilateral $PQRS$ formed from two triangles, PQS and QRS.

Calculate

(i) QR,

$QR =$ ..... cm [3]

(ii) PS,

$PS =$ ..... cm [3]

(iii) the area of quadrilateral $PQRS$.

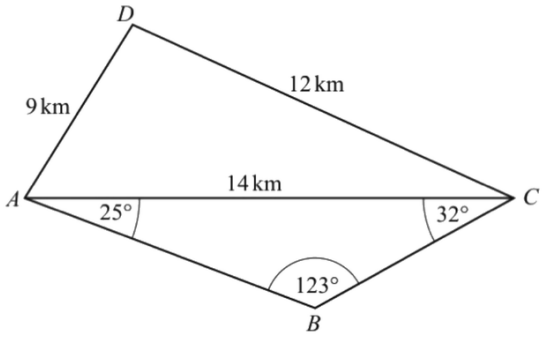
..... cm^2 [4]

75 (i) $\dfrac{QR}{\sin 25^\circ} = \dfrac{6}{\sin 72^\circ} \Rightarrow QR = \boxed{2.67 \text{ cm}}$

(ii) $PS^2 = 7.4^2 + 6^2 - 2(7.4)(6)\cos 34^\circ \Rightarrow PS = \boxed{4.14 \text{ cm}}$

(iii) $\angle SQR = 83^\circ$. Area $= \frac{1}{2}(7.4)(6)\sin 34^\circ + \frac{1}{2}(6)(2.666)\sin 83^\circ = 12.41 + 7.94 =$ $\boxed{20.4 \text{ cm}^2}$

76



NOT TO SCALE

(a) Calculate angle ACD.

Angle ACD = ... [4]

(b) Show that $BC = 7.05\,\text{km}$, correct to 2 decimal places.

(c) Calculate the shortest distance from B to AC.

..... km [3]

(d) Calculate the length of the straight line BD.

BD = ... km [4]

(e) C is due east of A.

Find the bearing of D from C.

..... [2]

76 (a) $\cos ACD = \dfrac{14^2 + 12^2 - 9^2}{2(14)(12)} = 0.7708 \Rightarrow \boxed{39.6^\circ}$

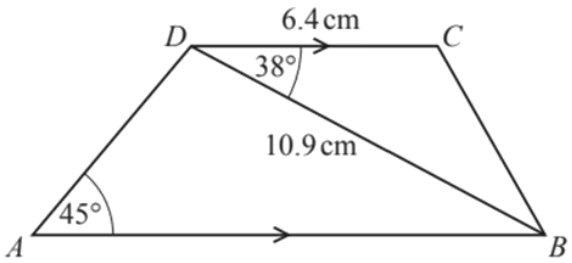
(b) $\dfrac{BC}{\sin 25^\circ} = \dfrac{14}{\sin 123^\circ} \Rightarrow BC = 7.0548 = 7.05$ km ✓

(c) $7.055 \sin 32^\circ = \boxed{3.74 \text{ km}}$

(d) $\angle BCD = 39.57^\circ + 32^\circ = 71.57^\circ$; $BD^2 = 7.055^2 + 12^2 - 2(7.055)(12)\cos 71.57^\circ \Rightarrow$ $BD = \boxed{11.8 \text{ km}}$

(e) C is due east of A, so from C the direction of A is 270°; D is 39.6° further round towards north: $270 + 39.6 = \boxed{309.6^\circ}$ (310°)

77



NOT TO SCALE

ABCD is a trapezium with *DC* parallel to *AB*.
DC = 6.4 cm, *DB* = 10.9 cm, angle *CDB* = 38° and angle *DAB* = 45°.

(a) Find *CB*.

CB = ... cm [3]

(b) **(i)** Find angle *ADB*.

Angle *ADB* = ... [1]

(ii) Find *AB*.

AB = ... cm [3]

(c) Calculate the area of the trapezium.

..... cm^2 [3]

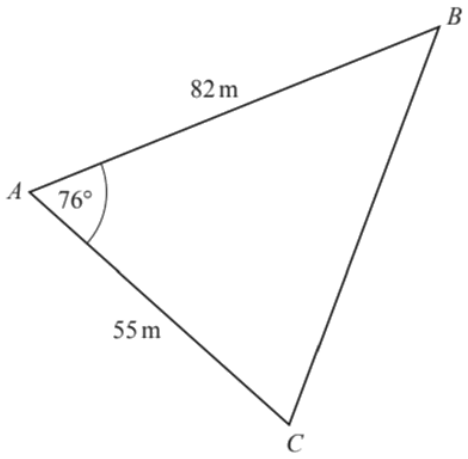
77 (a) $CB^2 = 6.4^2 + 10.9^2 - 2(6.4)(10.9)\cos 38^\circ \Rightarrow CB = \boxed{7.06 \text{ cm}}$

(b)(i) $\angle ABD = 38^\circ$ (alternate), so $\angle ADB = 180 - 45 - 38 = \boxed{97^\circ}$

(ii) $AB = \dfrac{10.9 \sin 97^\circ}{\sin 45^\circ} = \boxed{15.3 \text{ cm}}$

(c) Height $= 10.9 \sin 38^\circ = 6.711$; area $= \frac{1}{2}(6.4 + 15.30) \times 6.711 = \boxed{72.8 \text{ cm}^2}$

78



NOT TO SCALE

The diagram shows a field ABC.

(a) Calculate BC.

$BC =$ ..... m [3]

(b) Calculate angle ACB.

Angle $ACB =$ ..... [3]

(c) A gate, G, lies on AB at the shortest distance from C.

Calculate AG.

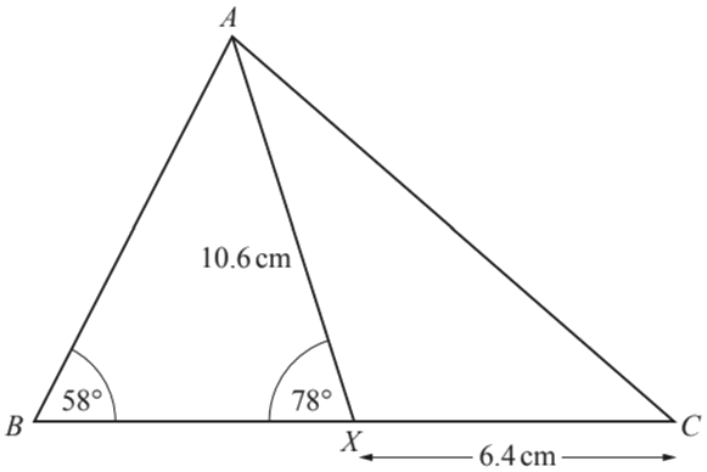
$AG =$ ..... m [3]

78 (a) $BC^2 = 82^2 + 55^2 - 2(82)(55)\cos 76° \Rightarrow BC = \boxed{87.0 \text{ m}}$

(b) $\sin C = \dfrac{82 \sin 76°}{86.99} \Rightarrow \boxed{66.2°}$

(c) $AG = 55 \cos 76° = \boxed{13.3 \text{ m}}$

79



NOT TO SCALE

The diagram shows triangle ABC.
X is a point on BC.
$AX = 10.6$ cm, $XC = 6.4$ cm, angle $ABC = 58°$ and angle $AXB = 78°$.

(a) Calculate AC.

$AC =$ ..... cm [4]

(b) Calculate BX.

$BX =$ ..... cm [4]

(c) Calculate the area of triangle ABC.

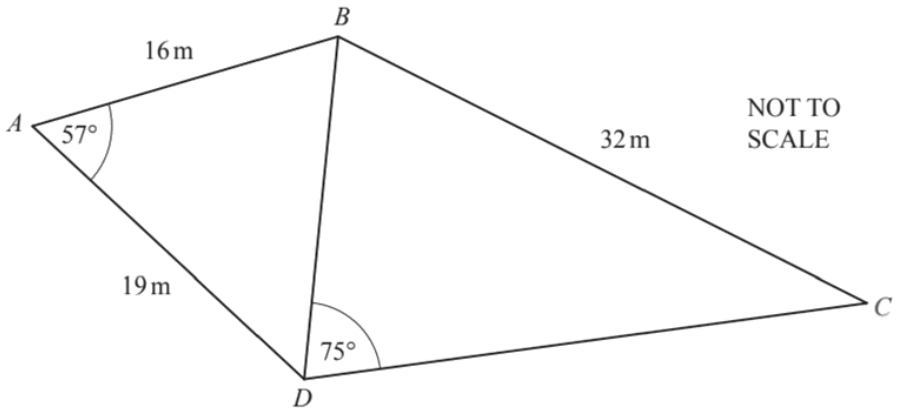
..... cm^2 [3]

79 (a) $\angle AXC = 102°$: $AC^2 = 10.6^2 + 6.4^2 - 2(10.6)(6.4)\cos 102° \Rightarrow AC = \boxed{13.5 \text{ cm}}$

(b) $\angle BAX = 180 - 58 - 78 = 44°$: $BX = \dfrac{10.6\sin 44°}{\sin 58°} = \boxed{8.68 \text{ cm}}$

(c) $\frac{1}{2}(8.683 + 6.4)(10.6)\sin 78° = \boxed{78.2 \text{ cm}^2}$

80



The diagram shows a quadrilateral *ABCD* made from two triangles, *ABD* and *BCD*.

(a) Show that $BD = 16.9$ m, correct to 1 decimal place.

[3]

(b) Calculate angle *CBD*.

Angle *CBD* = ... [4]

(c) Find the area of the quadrilateral *ABCD*.

..... m^2 [3]

(d) Find the shortest distance from *B* to *AD*.

..... m [3]

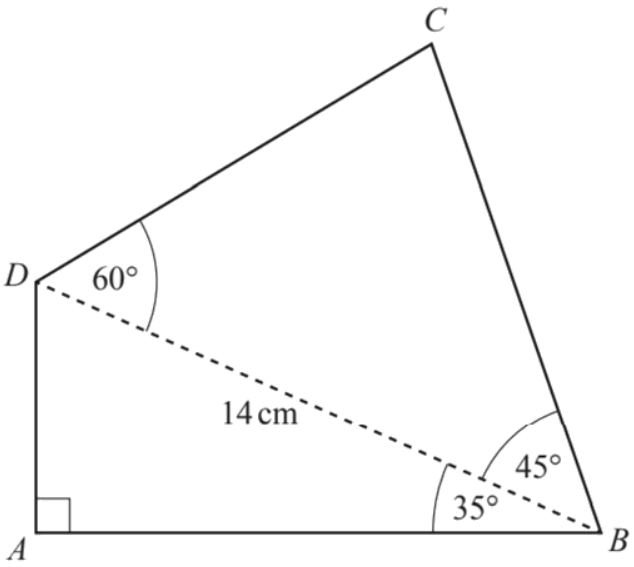
80 (a) $BD^2 = 16^2 + 19^2 - 2(16)(19)\cos 57^\circ = 285.86 \Rightarrow BD = 16.9$ m ✓

(b) $\sin C = \dfrac{16.91 \sin 75^\circ}{32} \Rightarrow C = 30.7^\circ$; $\angle CBD = 180 - 75 - 30.7 = \boxed{74.3^\circ}$

(c) $\frac{1}{2}(16)(19)\sin 57^\circ + \frac{1}{2}(16.91)(32)\sin 74.3^\circ = 127.5 + 260.4 = \boxed{388 \text{ m}^2}$

(d) $16 \sin 57^\circ = \boxed{13.4 \text{ m}}$

81



NOT TO SCALE

Calculate the perimeter of the quadrilateral $ABCD$.

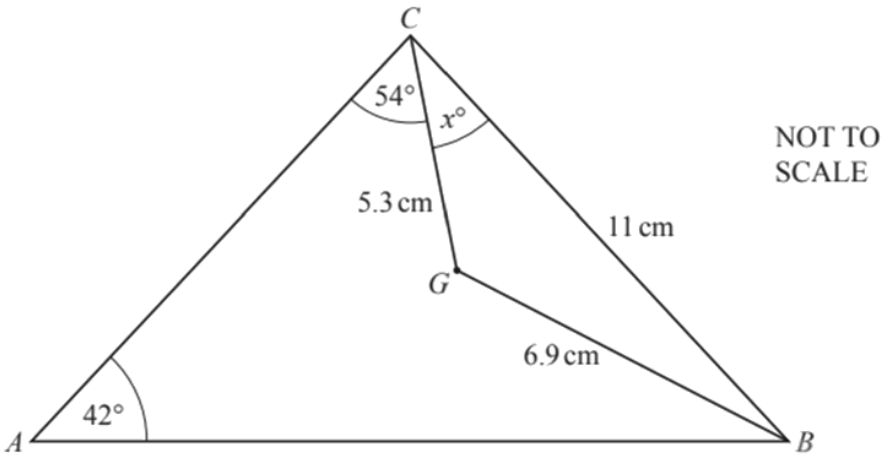
..... cm [7]

81 $AD = 14\sin 35^\circ = 8.030,\ AB = 14\cos 35^\circ = 11.468.\ \angle C = 180 - 60 - 45 = 75^\circ$

$DC = \dfrac{14\sin 45^\circ}{\sin 75^\circ} = 10.249,\ BC = \dfrac{14\sin 60^\circ}{\sin 75^\circ} = 12.552$

Perimeter $= 8.030 + 11.468 + 12.552 + 10.249 =$ $\boxed{42.3 \text{ cm}}$

82



The diagram shows triangle ABC with point G inside.
$CB = 11$ cm, $CG = 5.3$ cm and $BG = 6.9$ cm.
Angle $CAB = 42°$ and angle $ACG = 54°$.

(i) Calculate the value of x.

$x =$ ..... [4]

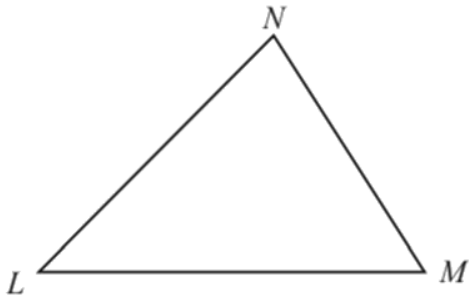
(ii) Calculate AC.

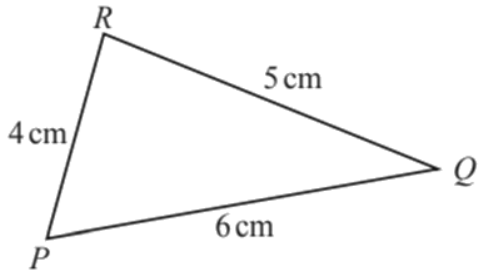
$AC =$ ..... cm [4]

82 (i) $\cos x = \dfrac{5.3^2 + 11^2 - 6.9^2}{2(5.3)(11)} = 0.8704 \Rightarrow \boxed{x = 29.5}$

(ii) $\angle ACB = 54 + 29.5 = 83.5°$, $\angle B = 180 - 42 - 83.5 = 54.5°$. $AC = \dfrac{11\sin 54.5°}{\sin 42°} =$ $\boxed{13.4 \text{ cm}}$

83





NOT TO SCALE

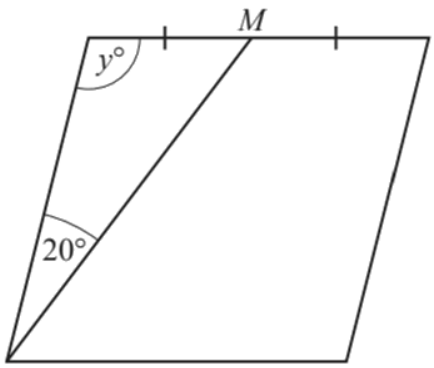
In triangle LMN, the ratio angle L : angle M : angle $N = 4 : 5 : 6$.
In triangle PQR, $PQ = 6\,\text{cm}$, $PR = 4\,\text{cm}$ and $QR = 5\,\text{cm}$.

Calculate the difference between the largest angle in triangle PQR and the largest angle in triangle LMN.

..... [7]

83 LMN: $4+5+6=15$ parts, $180\div 15=12$, so angles $48^\circ, 60^\circ, 72^\circ$ — largest 72°. PQR: largest angle opposite 6: $\cos R = \dfrac{4^2+5^2-6^2}{2(4)(5)} = \frac{1}{8} \Rightarrow 82.8^\circ$. Difference $=$ $\boxed{10.8^\circ}$

84



NOT TO SCALE

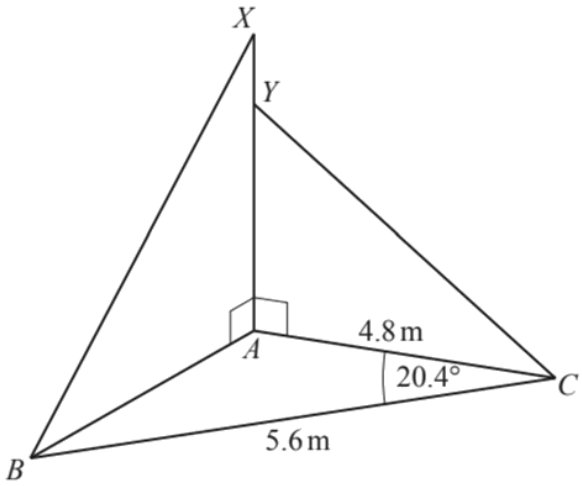
This rhombus has perimeter 20 cm and angle y is obtuse.
M is the midpoint of one of the sides.

Find the value of y.

$y =$ ..... [5]

84 Side $= 20 \div 4 = 5$; half side $= 2.5$. In the small triangle: $\dfrac{\sin\theta}{5} = \dfrac{\sin 20^\circ}{2.5} \Rightarrow \theta = 43.2^\circ$ (opposite 5)
$y = 180 - 20 - 43.2 = \boxed{116.8^\circ}$

85



NOT TO SCALE

ABC is a scalene triangle on horizontal ground.
AYX is a straight vertical post, held in place by two straight wires XB and YC.
$AC = 4.8$ m, $BC = 5.6$ m and angle $ACB = 20.4°$.

(i) Calculate AB.

$AB =$ ..... m [3]

(ii) Angle $XBA = 64°$.

Calculate AX.

$AX =$ ..... m [2]

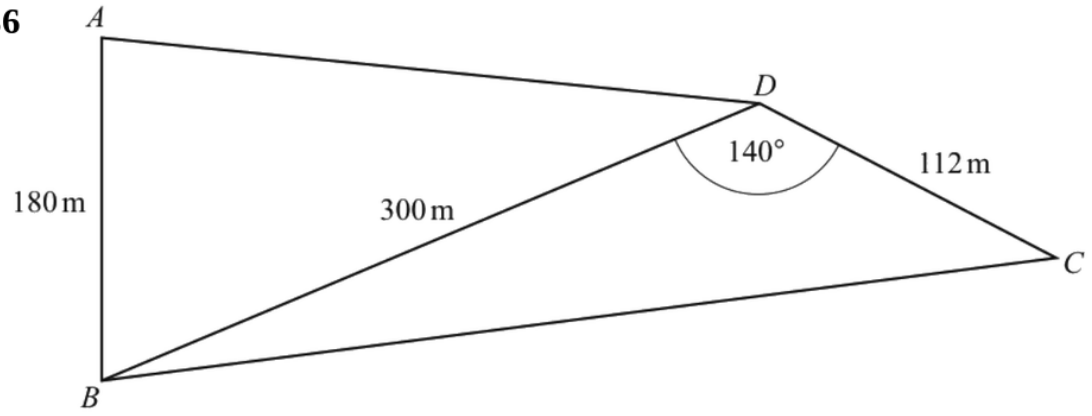
(iii) $AY = 2.9$ m.

Calculate the area of triangle YAC.

..... m^2 [2]

85 (i) $AB^2 = 4.8^2 + 5.6^2 - 2(4.8)(5.6)\cos 20.4^\circ \Rightarrow AB = \boxed{2.00\text{ m}}$
(ii) $AX = AB\tan 64^\circ = \boxed{4.11\text{ m}}$ (iii) Right angle at A: $\frac{1}{2} \times 4.8 \times 2.9 = \boxed{6.96\text{ m}^2}$

86



NOT TO SCALE

The diagram shows a field, *ABCD*, in the shape of a quadrilateral.
BD is a straight path across the field.

(a) Calculate *BC*.

BC = ... m [3]

(b) Calculate angle *DBC*.

Angle *DBC* = ... [3]

(c) The total area of the field, *ABCD*, is 35 900 m^2.

Work out the length of the shortest distance from *D* to *AB*.

..... m [4]

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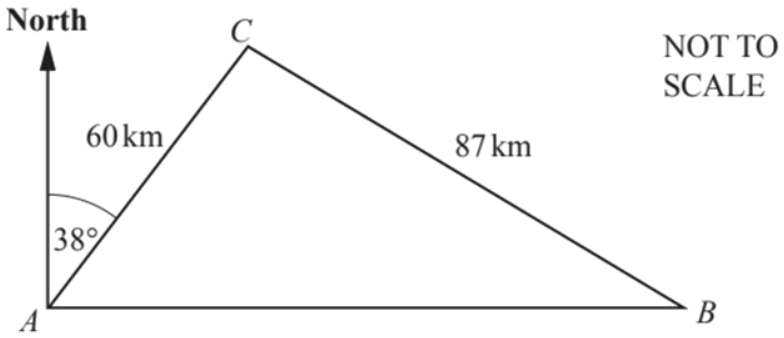
(a) $BC^2 = 300^2 + 112^2 - 2(300)(112)\cos 140° = 154\,022 \Rightarrow BC = \boxed{392\text{ m}}$

(b) $\dfrac{\sin DBC}{112} = \dfrac{\sin 140°}{392.46} \Rightarrow \angle DBC = \boxed{10.6°}$

(c) Area $BCD = \frac{1}{2}(112)(300)\sin 140° = 10\,798.8$; area $ABD = 35\,900 - 10\,798.8 = 25\,101.2$

Shortest distance d from D to AB is perpendicular: $\frac{1}{2} \times 180 \times d = 25\,101.2 \Rightarrow d = \boxed{279\text{ m}}$

87



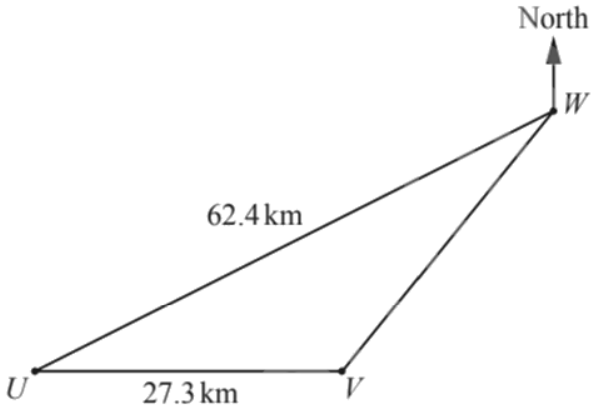
The diagram shows the straight roads between town A, town B and town C.
$AC = 60$ km, $CB = 87$ km and B is due east of A.
The bearing of C from A is 038°.

(a) Show that angle $ACB = 95.1°$, correct to 1 decimal place.

[5]

87 $\angle CAB = 90 - 38 = 52°$. $\sin B = \dfrac{60 \sin 52°}{87} \Rightarrow \angle ABC = 32.9°$
$\angle ACB = 180 - 52 - 32.9 = 95.1°$ ✓

88



NOT TO SCALE

The diagram shows the position of three towns, U, V and W.
U is due west of V and angle $UVW = 125°$.

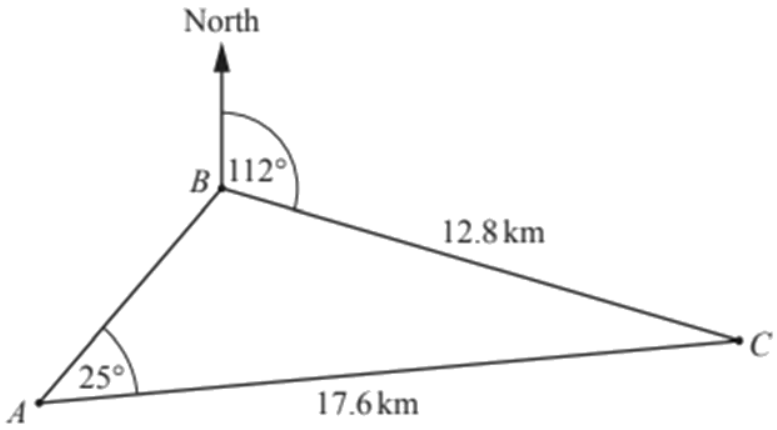
Calculate the bearing of U from W.

..... [4]

88 Sine rule: $\sin W = \dfrac{27.3\sin 125^\circ}{62.4} \Rightarrow \angle UWV = 21.0^\circ$, so $\angle WUV = 180 - 125 - 21.0 = 34.0^\circ$.

Bearing of W from $U = 90 - 34.0 = 056^\circ$, so bearing of U from $W = 056 + 180 = \boxed{236^\circ}$

89



NOT TO SCALE

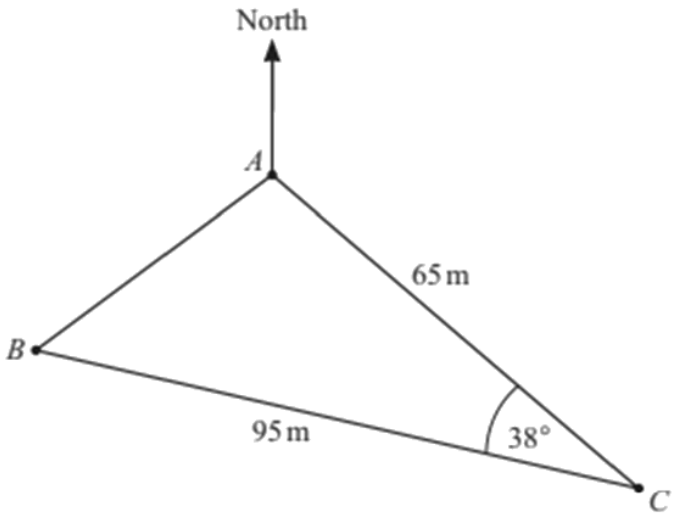
The diagram shows the positions of three ships A, B and C.
$AC = 17.6$ km, $BC = 12.8$ km and angle $BAC = 25°$.
The bearing of C from B is 112° and angle ABC is obtuse.

Calculate the bearing of B from A.

..... [5]

89 $\sin B = \dfrac{17.6\sin 25^\circ}{12.8} = 0.5811 \Rightarrow 35.5^\circ$; obtuse so $\angle ABC = 144.5^\circ$.
Bearing of A from $B = 112 + 144.5 = 256.5^\circ$, so bearing of B from $A = 256.5 - 180 =$ $\boxed{076.5^\circ}$

90



NOT TO SCALE

The diagram shows three points A, B and C on horizontal ground.
The bearing of C from A is 145° and angle $ACB = 38°$.
$AC = 65$ m and $BC = 95$ m.

(a) Find the bearing of B from C.

..... [2]

(b) Show that $AB = 59.3$ m, correct to 1 decimal place.

[3]

(c) Angle BAC is obtuse.

Work out the bearing of B from A.

..... [4]

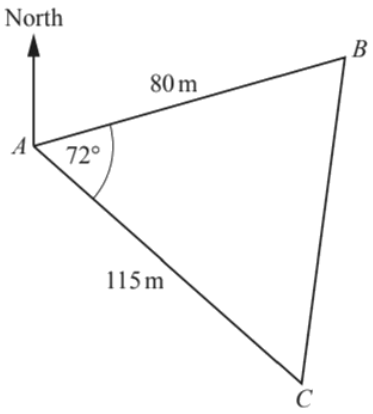
90 (a) Bearing of A from $C = 325°$; B is $38°$ further anticlockwise: $325 - 38 = \boxed{287°}$

(b) $AB^2 = 65^2 + 95^2 - 2(65)(95)\cos 38° \Rightarrow AB = 59.31 = 59.3$ m ✓

(c) $\sin A = \dfrac{95 \sin 38°}{59.31} = 0.9861 \Rightarrow 80.4°$; angle BAC is obtuse $\Rightarrow 99.6°$.

Bearing of B from $A = 145 + 99.6 = \boxed{244.6°}$

91



NOT TO SCALE

The diagram shows the positions of three points A, B and C in a field.

(a) Show that BC is 118.1 m, correct to 1 decimal place.

[3]

(b) Calculate angle ABC.

Angle ABC = ... [3]

(c) The bearing of C from A is 147°.

Find the bearing of

(i) A from B,

..... [3]

(ii) B from C.

..... [2]

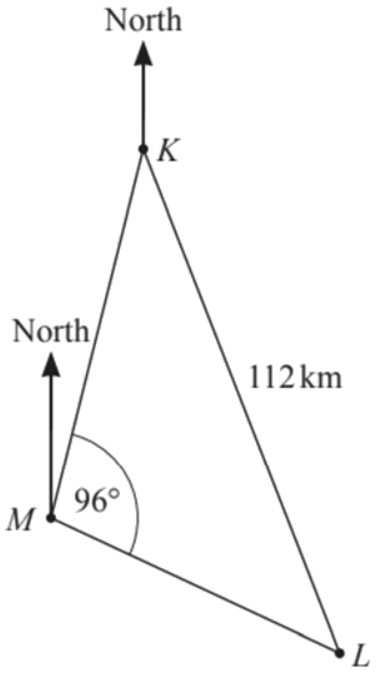
91 (a) $BC^2 = 80^2 + 115^2 - 2(80)(115)\cos 72° \Rightarrow BC = 118.06 = 118.1$ m ✓

(b) $\sin B = \dfrac{115 \sin 72°}{118.06} \Rightarrow \boxed{67.9°}$

(c)(i) Bearing of B from $A = 147 - 72 = 075°$, so A from $B = \boxed{255°}$

(ii) $\angle ACB = 180 - 72 - 67.9 = 40.1°$; A from $C = 327°$; B from $C = 327 + 40.1 - 360 = \boxed{007°}$

92



NOT TO SCALE

The diagram shows the positions of a lighthouse, L, and two ships, K and M.
The bearing of L from K is 155° and $KL = 112\,\text{km}$.
The bearing of K from M is 010° and angle $KML = 96°$.

Find the bearing and distance of ship M from the lighthouse, L.

Bearing .....

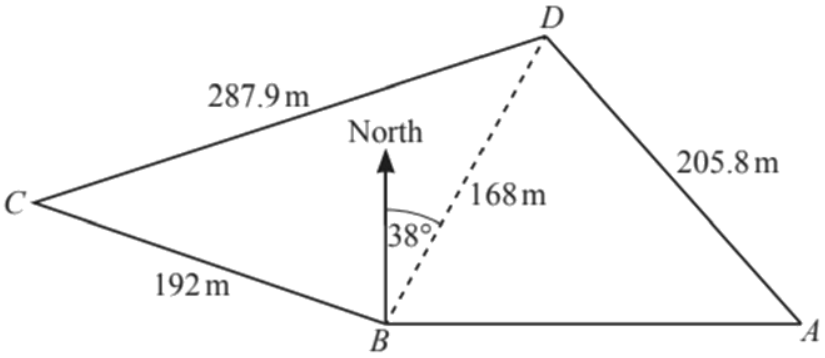
Distance ..... km [5]

92 At K: bearing to $M = 190°$, to $L = 155°$, so $\angle MKL = 35°$; $\angle KLM = 180 - 96 - 35 = 49°$.

Bearing of K from $L = 335°$, so M from $L = 335 - 49 = \boxed{286°}$

$ML = \dfrac{112 \sin 35°}{\sin 96°} = \boxed{64.6 \text{ km}}$

93



NOT TO SCALE

The diagram shows a field, $ABCD$, on horizontal ground.
$BC = 192$ m, $CD = 287.9$ m, $BD = 168$ m and $AD = 205.8$ m.

(a) (i) Calculate angle CBD and show that it rounds to 106.0°, correct to 1 decimal place.

[4]

(ii) The bearing of D from B is 038°.

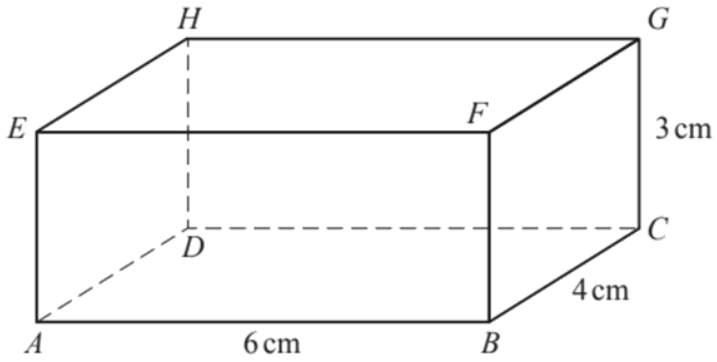
Find the bearing of C from B.

..... [1]

93 (i) $\cos CBD = \dfrac{192^2 + 168^2 - 287.9^2}{2(192)(168)} = -0.2759 \Rightarrow 106.0^\circ$ ✓

(ii) C is 106° anticlockwise from D: $38 - 106 = -68 \Rightarrow \boxed{292^\circ}$

94



NOT TO SCALE

ABCDEFGH is a cuboid.

Find the length of *AG*.
Give your answer in surd form.

..... cm [3]

95 A cuboid measures 24 cm by 12 cm by 8 cm.

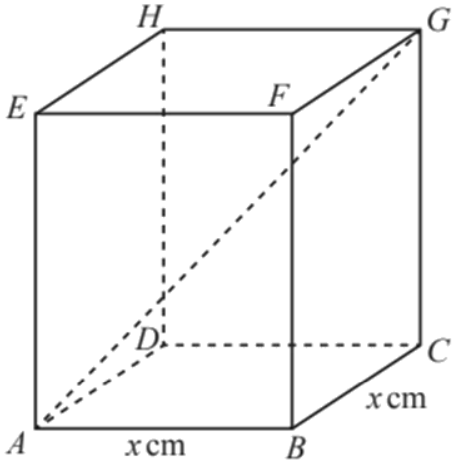
Calculate the length of a diagonal of the cuboid.

..... cm [3]

94 $AG = \sqrt{6^2 + 4^2 + 3^2} = \boxed{\sqrt{61}\text{ cm}}$

95 $\sqrt{24^2 + 12^2 + 8^2} = \sqrt{784} = \boxed{28\text{ cm}}$

96



NOT TO SCALE

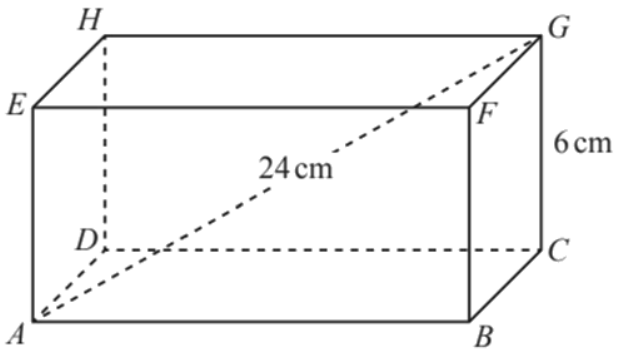
ABCDEFGH is a cuboid with a square base of side x cm.
$CG = 20$ cm and $AG = 28$ cm.

Calculate the value of x.

$x =$ ..... [4]

96 $x^2 + x^2 + 20^2 = 28^2 \Rightarrow 2x^2 = 384 \Rightarrow x^2 = 192 \Rightarrow \boxed{x = 13.9}$ $(8\sqrt{3})$

97



NOT TO SCALE

The diagram shows a cuboid $ABCDEFGH$.
$CG = 6$ cm, $AG = 24$ cm and $AB = 2BC$.

(a) Calculate AB.

$AB =$ ..... cm [4]

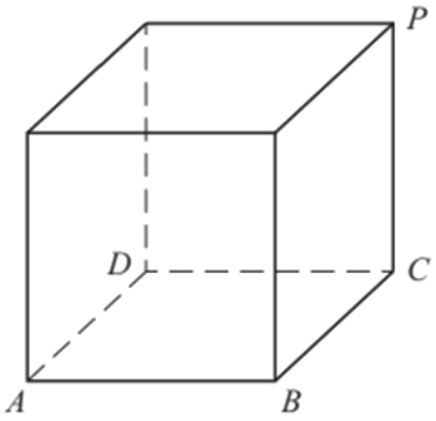
(b) Calculate the angle between AG and the base $ABCD$.

..... [3]

97 (a) $AC^2 = 24^2 - 6^2 = 540$; $(2x)^2 + x^2 = 540 \Rightarrow x^2 = 108 \Rightarrow BC = 10.39$, $AB = \boxed{20.8\text{ cm}}$

(b) $\sin\theta = \dfrac{6}{24} \Rightarrow \boxed{14.5^\circ}$

98



NOT TO SCALE

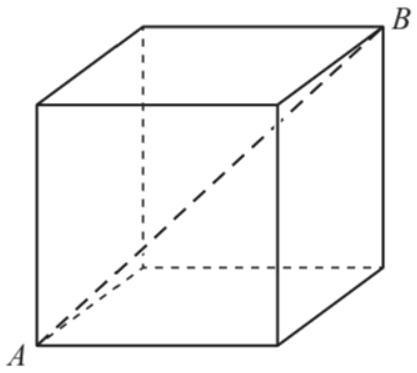
The diagram shows a cube.

Calculate the angle between the diagonal AP and the base $ABCD$.

..... [4]

98 Let edge $= 1$: $AC = \sqrt{2}$, $CP = 1$. $\tan\theta = \dfrac{1}{\sqrt{2}} \Rightarrow \boxed{35.3^\circ}$

99



NOT TO SCALE

The diagram shows a cube.
The length of the diagonal AB is 8.5 cm.

(i) Calculate the length of an edge of the cube.

..... cm [3]

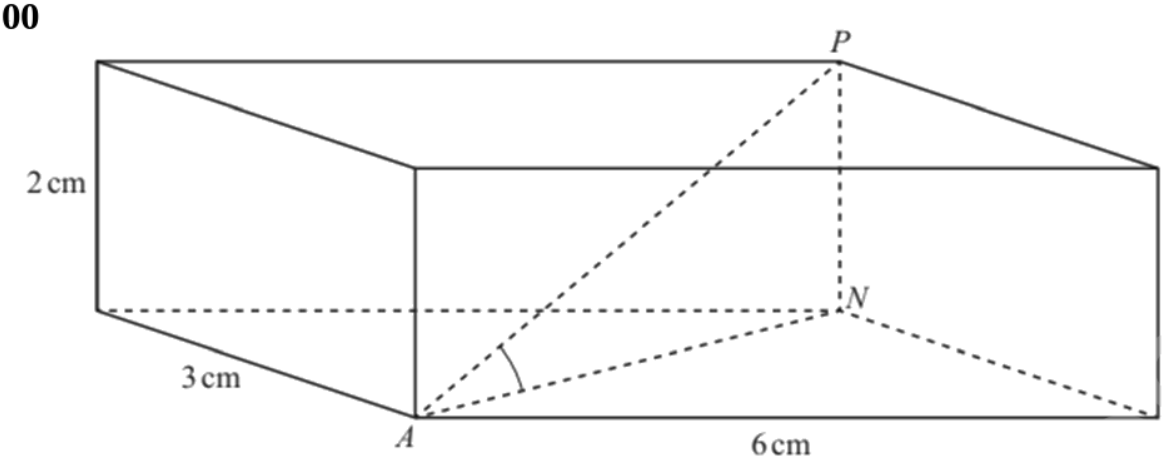
(ii) Calculate the angle between AB and the base of the cube.

..... [3]

99 (i) $3x^2 = 8.5^2 \Rightarrow x = \dfrac{8.5}{\sqrt{3}} = \boxed{4.91 \text{ cm}}$

(ii) $\sin\theta = \dfrac{4.907}{8.5} \Rightarrow \boxed{35.3^\circ}$

100



NOT TO SCALE

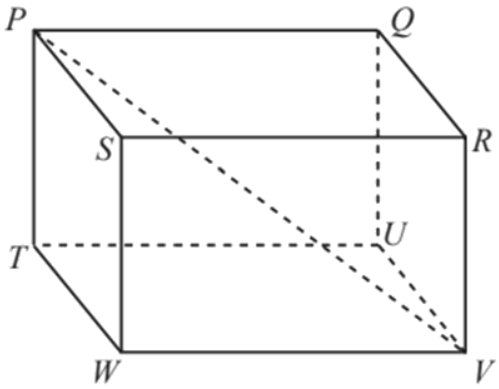
The diagram shows a cuboid measuring 6 cm by 3 cm by 2 cm.

Find the sine of the angle *PAN*, the angle between the diagonal *PA* and the base of the cuboid.

Sine of angle *PAN* = ... [4]

100 $AN = \sqrt{6^2+3^2} = \sqrt{45}$; $PA = \sqrt{45+2^2} = 7$. $\sin PAN = \dfrac{PN}{PA} = \boxed{\dfrac{2}{7}} \Rightarrow \angle PAN = \sin^{-1}\left(\frac{2}{7}\right) = \boxed{16.6^\circ}$

101



NOT TO SCALE

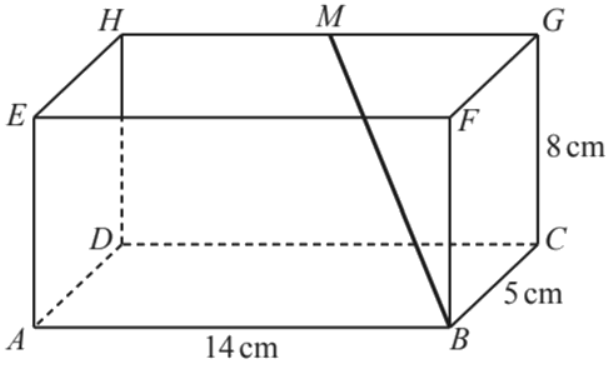
The diagram shows a cuboid $PQRSTUVW$.
$PV = 17.2\,\text{cm}$
The angle between the line PV and the base $TUVW$ of the cuboid is $43°$.

Calculate PT.

$PT =$ ..... cm [3]

101 Triangle PTV is right-angled at T: $PT = 17.2\sin 43^\circ = \boxed{11.7\text{ cm}}$

102



NOT TO SCALE

The diagram shows a cuboid $ABCDEFGH$.
$AB = 14$ cm, $BC = 5$ cm and $CG = 8$ cm.
M is the midpoint of HG.

(a) Calculate BM.

..... cm [3]

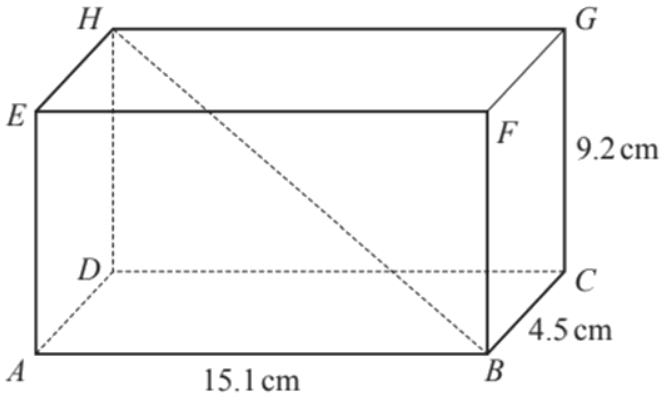
(b) Calculate the angle that BM makes with the base $ABCD$.

..... [3]

102 (a) Drop M to N on DC: $NC = 7$; $NB = \sqrt{7^2 + 5^2}$; $BM = \sqrt{5^2 + 7^2 + 8^2} = \sqrt{138} = \boxed{11.7\text{ cm}}$

(b) $\sin\theta = \dfrac{8}{\sqrt{138}} \Rightarrow \boxed{42.9^\circ}$

103



NOT TO SCALE

The diagram shows a cuboid $ABCDEFGH$.
$AB = 15.1$ cm, $BC = 4.5$ cm and $CG = 9.2$ cm.

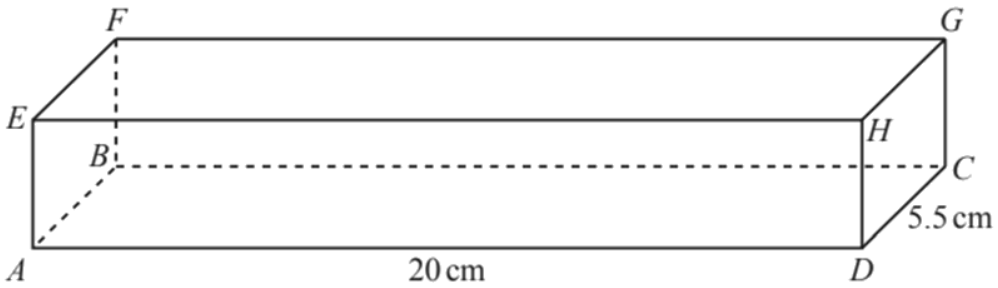
Calculate the angle that the diagonal BH makes with the face $ADHE$.

..... [4]

103 BA is perpendicular to face $ADHE$, so the angle is $\angle BHA$.

$AH = \sqrt{4.5^2 + 9.2^2} = 10.24$; $\tan\theta = \dfrac{15.1}{10.24} \Rightarrow \boxed{55.9^\circ}$

104



NOT TO SCALE

The diagram shows cuboid *ABCDEFGH* of length 20 cm and width 5.5 cm.
The volume of the cuboid is 495 cm^3.

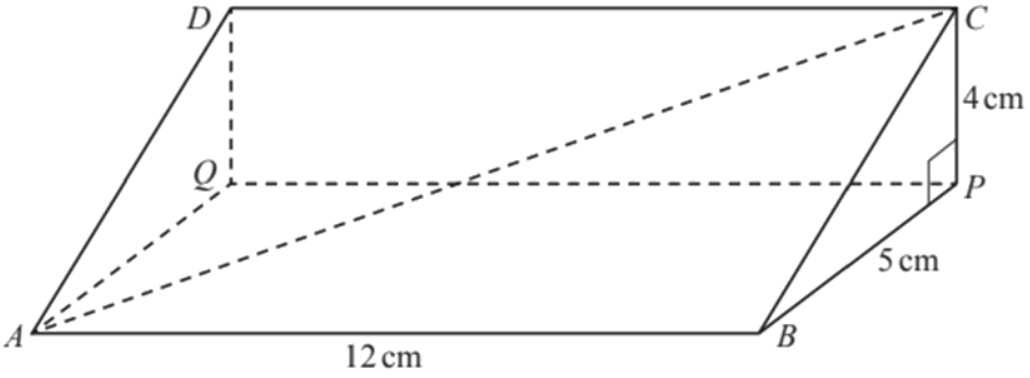
Find the angle between the line *AG* and the base of the cuboid *ABCD*.

..... [5]

104 $\text{Height} = \dfrac{495}{20 \times 5.5} = 4.5.$ $AC = \sqrt{20^2 + 5.5^2} = 20.74$

$\tan\theta = \dfrac{4.5}{20.74} \Rightarrow \boxed{12.2^\circ}$

105



NOT TO SCALE

The diagram shows a triangular prism.
Angle $BPC = 90°$.

(a) Calculate AC.

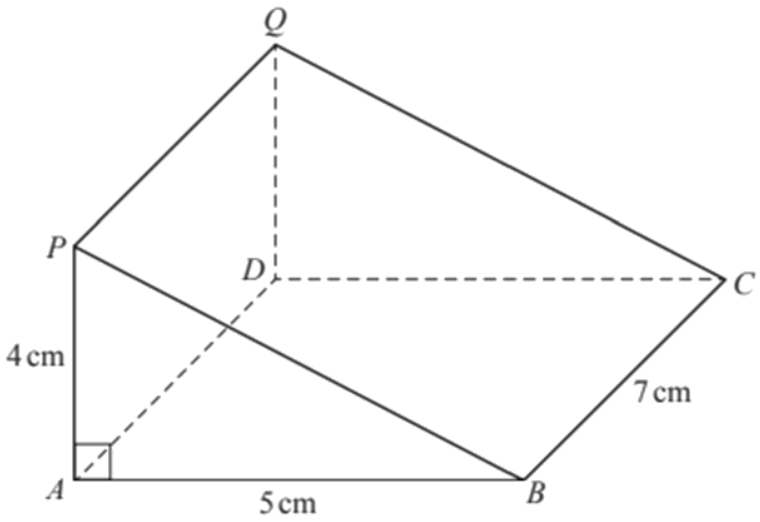
$AC =$ ..... cm [3]

(b) Calculate the angle between AC and the base $ABPQ$.

..... [3]

105 (a) $BC = \sqrt{5^2+4^2} = \sqrt{41}$; $AC = \sqrt{12^2+41} = \sqrt{185} = \boxed{13.6\text{ cm}}$
(b) $AP = \sqrt{12^2+5^2} = 13$, $CP = 4$: $\tan\theta = \frac{4}{13} \Rightarrow \boxed{17.1^\circ}$

106



NOT TO SCALE

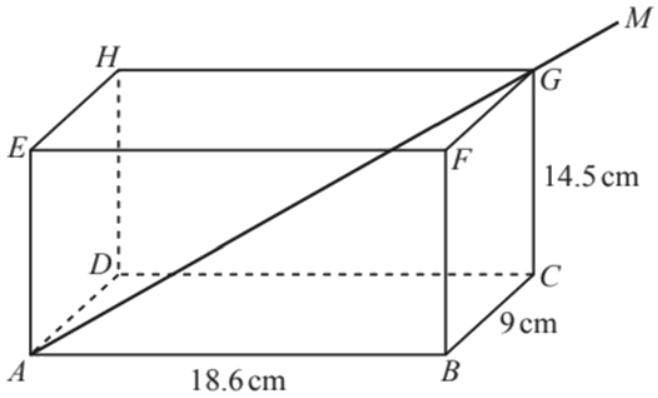
The diagram shows a triangular prism $ABCDQP$ of length 7 cm.
The cross-section is triangle PAB with $PA = 4$ cm, $AB = 5$ cm and angle $PAB = 90°$.

Calculate the angle between the line PC and the base $ABCD$.

..... [4]

106 $AC = \sqrt{5^2 + 7^2} = \sqrt{74}$. $\tan\theta = \dfrac{PA}{AC} = \dfrac{4}{\sqrt{74}} \Rightarrow \boxed{24.9^\circ}$

107



NOT TO SCALE

The diagram shows an open rectangular box $ABCDEFGH$.
$AB = 18.6$ cm, $BC = 9$ cm and $CG = 14.5$ cm.
A straight stick AGM rests against A and G and extends outside the box to M.

(a) Calculate the angle between the stick and the base of the box.

..... [4]

(b) $AM = 30$ cm.

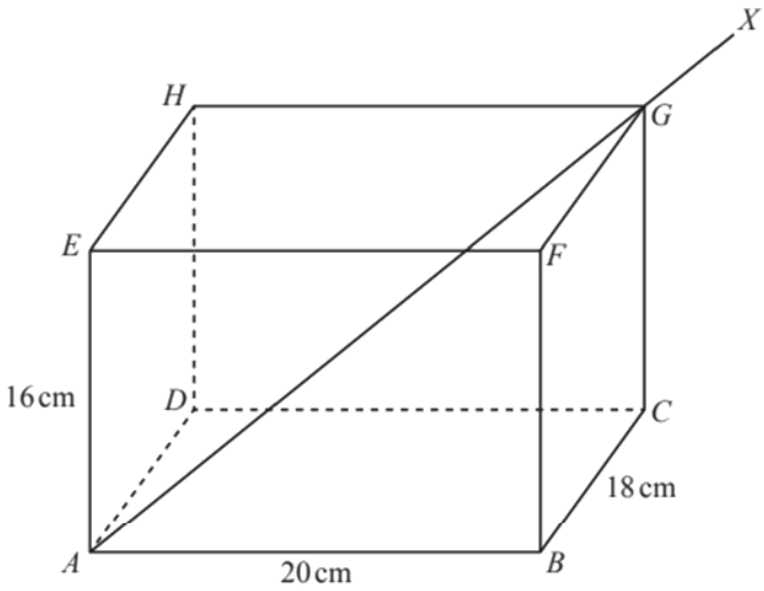
Show that $GM = 4.8$ cm, correct to 1 decimal place.

[3]

107 (a) $AC = \sqrt{18.6^2 + 9^2} = 20.66$; $\tan\theta = \dfrac{14.5}{20.66} \Rightarrow \boxed{35.1^\circ}$

(b) $AG = \sqrt{20.66^2 + 14.5^2} = 25.24$; $GM = 30 - 25.24 = 4.76 = 4.8$ cm ✓

108



NOT TO SCALE

The diagram shows an open box $ABCDEFGH$ in the shape of a cuboid.
$AB = 20$ cm, $BC = 18$ cm and $AE = 16$ cm.
A thin rod AGX rests partly in the box as shown.
The rod is 40 cm long.

(i) Calculate GX, the length of the rod which is outside the box.

$GX =$ ..... cm [4]

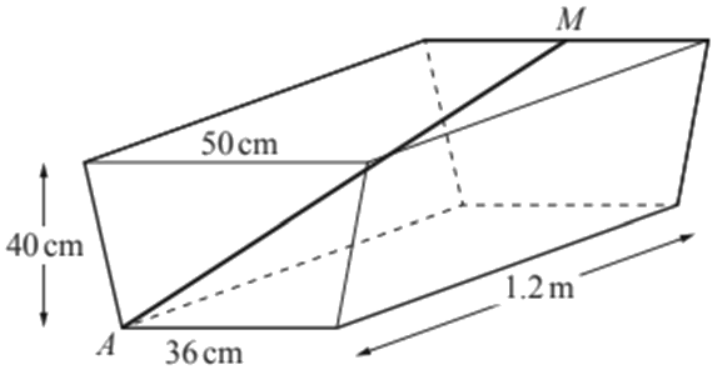
(ii) Calculate the angle the rod makes with the base of the box.

..... [3]

108 (i) $AG = \sqrt{20^2 + 18^2 + 16^2} = \sqrt{980} = 31.30$; $GX = 40 - 31.30 = \boxed{8.70 \text{ cm}}$

(ii) $\sin\theta = \dfrac{16}{31.30} \Rightarrow \boxed{30.7^\circ}$

109



NOT TO SCALE

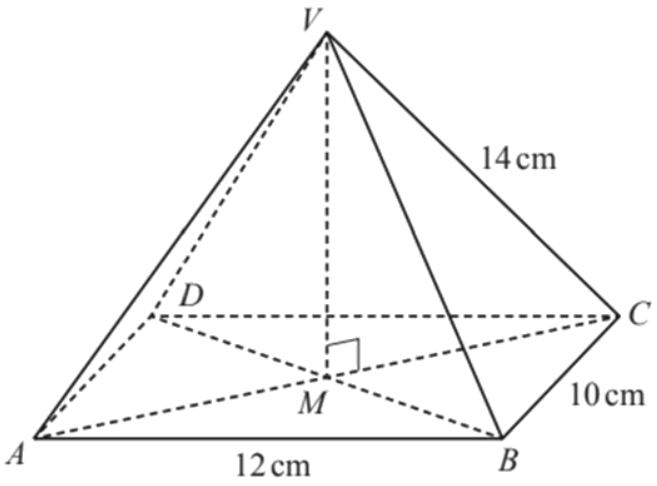
A steel rod AM is placed inside the empty water trough as shown in the diagram.
A is a vertex at the base of the isosceles trapezium and M is the midpoint of the top edge on the opposite face.

Calculate the length of the steel rod, AM.

$AM =$ ..... cm [4]

109 Horizontal offset across the trapezium $= \frac{36}{2} = 18$ cm (from corner to centre line), height 40, length 1.2 m $= 120$ cm.
$AM = \sqrt{18^2 + 40^2 + 120^2} = \sqrt{16\,324} = \boxed{128 \text{ cm}}$

110



NOT TO SCALE

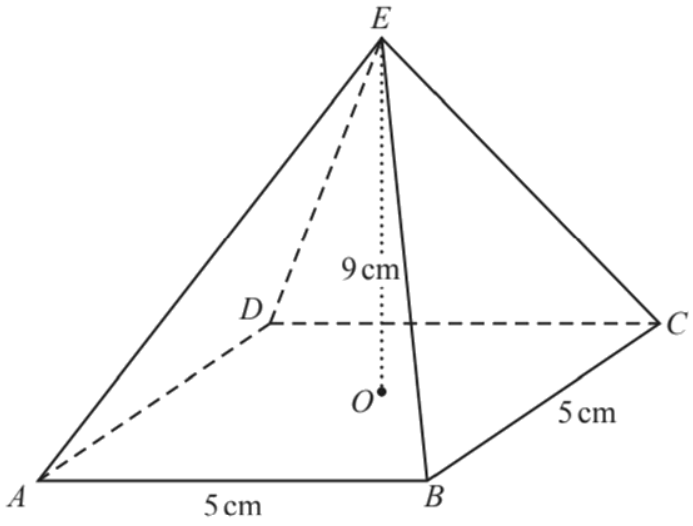
The diagram shows a pyramid $VABCD$ with a rectangular base.
V is vertically above M, the intersection of the diagonals AC and BD.
$AB = 12$ cm, $BC = 10$ cm and $VC = 14$ cm.

Calculate the angle that VC makes with the base $ABCD$.

..... [4]

110 $AC = \sqrt{12^2 + 10^2} = 15.62$, $MC = 7.810$. $\cos\theta = \dfrac{7.810}{14} \Rightarrow \boxed{56.1^\circ}$

111



NOT TO SCALE

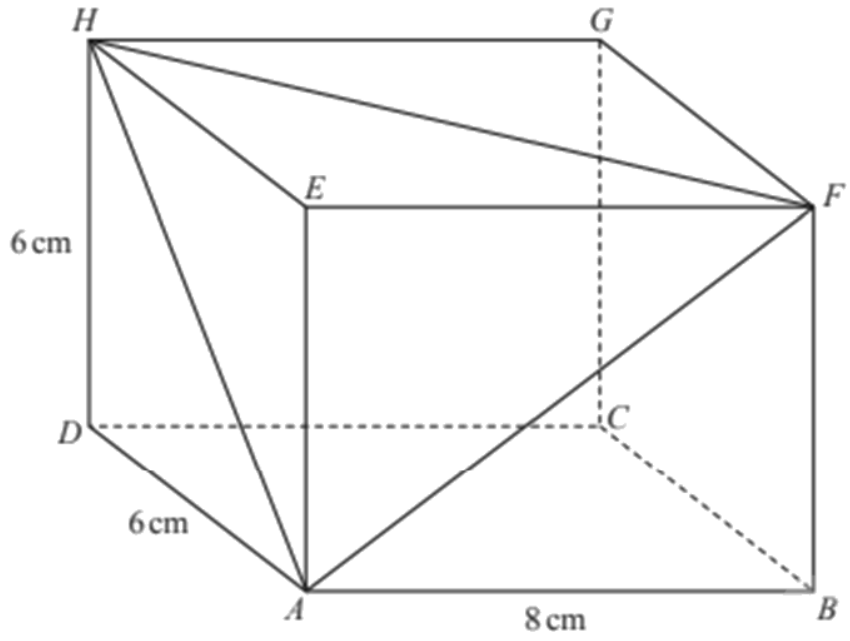
The diagram shows a pyramid *ABCDE*.
The pyramid has a square horizontal base *ABCD* with side 5 cm.
The vertex *E* is vertically above the centre *O* of the base.
The height *OE* of the pyramid is 9 cm.

Calculate the angle that *EC* makes with the base *ABCD*.

.....

111 $AC = \sqrt{5^2+5^2} = 7.071$, $OC = 3.536$. $\tan\theta = \dfrac{9}{3.536} \Rightarrow \boxed{68.6^\circ}$

112



NOT TO SCALE

The diagram shows a cuboid.
$AB = 8\text{cm}$, $AD = 6\text{cm}$ and $DH = 6\text{cm}$.

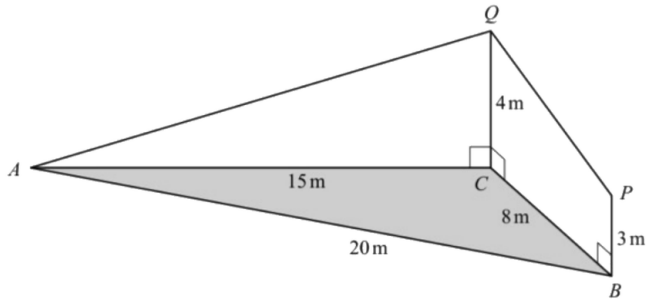
Calculate angle HAF.

Angle HAF = ... [6]

112 $HA = \sqrt{6^2+6^2} = \sqrt{72},\ AF = \sqrt{8^2+6^2} = 10,\ HF = \sqrt{8^2+6^2} = 10$

$$\cos HAF = \frac{72+100-100}{2\sqrt{72}(10)} = 0.4243 \Rightarrow \boxed{64.9^\circ}$$

113



NOT TO SCALE

The diagram shows triangle ABC on horizontal ground.
$AC = 15\,\text{m}$, $BC = 8\,\text{m}$ and $AB = 20\,\text{m}$.

BP and CQ are vertical poles of different heights.
$BP = 3\,\text{m}$ and $CQ = 4\,\text{m}$.
AQ and PQ are straight wires.

(a) Show that angle $ACB = 117.5°$, correct to 1 decimal place.

[4]

(b) Calculate the area of triangle ABC.

..... m^2 [2]

(c) Calculate the length of AQ.

..... m [2]

(d) Calculate the angle of elevation of Q from P.

..... [3]

(e) Another straight wire connects A to the midpoint of PQ.

Calculate the angle between this wire and the horizontal ground.

113 *Cambridge 0580 · Oct/Nov 2022 · Paper 41 · Q8*

(a) $\cos ACB = \dfrac{15^2+8^2-20^2}{2(15)(8)} = -\frac{111}{240} = -0.4625 \Rightarrow \angle ACB = 117.54^\circ = 117.5^\circ$ ✓

(b) $\frac{1}{2}(8)(15)\sin 117.5^\circ = \boxed{53.2\text{ m}^2}$ (c) $AQ = \sqrt{15^2+4^2} = \boxed{15.5\text{ m}}$

(d) $\tan\theta = \dfrac{4-3}{8} \Rightarrow \theta = \boxed{7.13^\circ}$

(e) Midpoint M of PQ is 3.5 m high, above the midpoint N of BC ($CN = 4$).

$AN^2 = 15^2+4^2-2(15)(4)\cos 117.5^\circ \Rightarrow AN = 17.22$; $\tan\phi = \dfrac{3.5}{17.22} \Rightarrow \phi = \boxed{11.5^\circ}$